Aspirin

Mechanism of action

- Aspirin (acetylsalicylic acid) has antiplatelet, antipyretic, anti-inflammatory and analgesic effects. In the out-of-hospital setting aspirin is only administered for its antiplatelet activity.
- Aspirin inhibits the enzyme cyclooxygenase which results in a reduction in the formation of prostaglandins and thromboxane.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to 'cardiac chest pain' in the FR Field Guide.

Indications

- ✓ Myocardial ischaemia.
- ✓ STEMI.

Contraindications

- ✕ Known severe allergy.
- ✕ Third trimester of pregnancy.

Cautions

- ⚠ Known bleeding disorder. Aspirin will increase the risk of bleeding, however the balance of risk is usually in favour of administering aspirin.
- ⚠ Clinically significant bleeding. Aspirin will increase the risk of bleeding and the nature and risk of the bleeding must be taken into account.
- ⚠ Known worsening of bronchospasm with NSAIDs. Some patients with asthma or COPD have known worsening of bronchospasm with NSAIDs (including aspirin) and a decision must be made based on the balance of risk. If there is a clear history of significant bronchospasm with NSAIDs, aspirin should be withheld.

Aspirin

Use in pregnancy or when breastfeeding

- May cause harm during pregnancy. Aspirin has been associated with premature delivery and premature closure of the ductus arteriosus, when administered in the third trimester of pregnancy.
- The likelihood of clinically important myocardial ischaemia occurring in a woman who is pregnant is so low that the balance of risk is usually in favour of aspirin being withheld.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 300 mg.

Administration

- Administer PO. Dispersible tablets may be chewed or dissolved in water.

Common adverse effects

- Increased bleeding.
- Although indigestion, gastrointestinal ulceration and gastrointestinal bleeding are commonly listed as adverse effects, these are only associated with long-term administration.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 3-5 days for the antiplatelet activity. This is because platelets exposed to aspirin are impaired for the life of the platelet which is 7-10 days. Approximately 10% of platelets are replaced each day.

Usual preparation

- 300 mg dispersible tablets.

Aspirin

Pharmacokinetics

- Absorption occurs in the stomach and small intestine. The presence of food in the stomach will delay absorption, but this is not usually clinically significant.
- Aspirin is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Aspirin displaces warfarin from binding sites and increases the activity of warfarin. However, this effect is most prominent with chronic administration and aspirin is indicated if a patient taking warfarin has clinically significant myocardial ischaemia.

Additional information

- 111 Call Handlers will give instructions for patients to take aspirin if they are suspected of having myocardial ischaemia. This means that some patients that do not have myocardial ischaemia will be advised to take aspirin. The risk for these patients is extremely low and there is potential benefit for those patients who are experiencing myocardial ischaemia.
- The contraindications to aspirin in the call handling protocols are different to those within these CPGs and these differences are not clinically significant.
- Many patients have enteric coated aspirin and not dispersible aspirin and they will be given instructions to chew and swallow their enteric coated aspirin.
- Enteric coated aspirin is not destroyed when it is chewed and will be absorbed. It is however, quite unpleasant to chew and swallow.
  - If the patient still has the tablets in their mouth, ask them to spit the tablets out and administer an additional 300 mg of dispersible aspirin.
  - If the patient has chewed and swallowed 300 mg of aspirin (including enteric coated aspirin), do not administer additional aspirin.
  - If the patient has swallowed (without chewing) 300 mg of enteric coated aspirin, administer an additional 300 mg of dispersible aspirin. This is because absorption of the enteric coated aspirin will be delayed.
  - If it is unclear what the patient has taken, administer an additional 300 mg of dispersible aspirin.
- Urea also impairs platelet function, but aspirin is not contraindicated in the setting of renal failure.

Aspirin

More Information

Version history

11 September 2023 1.0.4

Update: New indication to align with administrtaion for STEMI.

GTN (Glyceryl Trinitrate) Spray

Mechanism of action

- GTN is a vasodilator. It acts on vascular smooth muscle to cause venous and arterial vasodilation, with the predominant effect being on veins.
- The mechanism of action is not clear, but it appears that GTN results in the formation of nitric oxide which is a vasodilator. GTN causes:
  - A reduction in venous return (preload) to the heart. This reduces ventricular filling and cardiac output which reduces myocardial oxygen demand.
  - Arterial dilation which reduces peripheral resistance (afterload). This reduces the force the left ventricle must overcome to eject blood into the arteries which reduces myocardial oxygen demand.
  - Dilation of the coronary arteries which may increase coronary blood supply, though this is not usually clinically significant.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Myocardial ischaemia.
- ✓ Cardiogenic pulmonary oedema.
- ✓ Control of hypertension associated with autonomic dysreflexia.
- ✓ Control of hypertension (usually in conjunction with labetalol) prior to fibrinolytic treatment for STEMI.
- ✓ Control of hypertension (usually in conjunction with labetalol) during inter-hospital transfer for STEMI.
- ✓ STEMI

GTN (Glyceryl Trinitrate) Spray

Contraindications

- ✕ Known severe allergy.
- ✕ Systolic BP less than 110 mmHg.
- ✕ Heart rate less than 40/minute.
- ✕ Heart rate greater than 150/minute.
- ✕ Ventricular tachycardia.

Cautions

- ⚠ STEMI, particularly STEMI involving the right ventricle. GTN may cause a significant fall in cardiac output and if there are signs of low cardiac output GTN should be withheld.
- ⚠ The patient is frail.
- ⚠ Signs of shock. The patient's reduced cardiac output may fall further with GTN administration.
- ⚠ Dysrhythmia. This may cause a reduced cardiac output which may fall further with GTN administration.
- ⚠ Has taken a phosphodiesterase inhibitor in the last 24 hours. See common interactions.
- ⚠ Known aortic or mitral stenosis. With aortic or mitral stenosis, cardiac output may be reduced as a result of the narrowed valve and a fall in preload may cause a further fall in cardiac output.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring GTN is very low, but GTN should be administered if indicated.

Dosage

- Myocardial ischaemia: 0.4 mg every five minutes. Consider increasing the dosing interval to ten minutes if a caution is present.
- STEMI: 0.4 mg with caution. Withhold if signs of poor perfusion are present.
- Cardiogenic pulmonary oedema: 0.8 mg every five minutes. Consider increasing the dosing interval to ten minutes if a caution is present. Increase the dose and frequency if the patient is not improving.

GTN (Glyceryl Trinitrate) Spray

- Control of hypertension: 0.4-0.8 mg every five minutes as required.

Administration

- Spray under the tongue. If this cannot be achieved it is acceptable to spray into the mouth.
- If GTN is administered in the presence of a caution:
  - The patient should be lying flat.
  - IV access should have been obtained whenever possible.
  - The dosing interval should be increased to ten minutes.
  - Be ready to administer 0.9% sodium chloride IV if there is a significant fall in cardiac output or blood pressure.

Common adverse effects

- Hypotension.
- Flushing.
- Headache.
- Tachycardia.
- Feeling light-headed.

Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 15-30 minutes.

Usual preparation

- Metered dose bottle delivering 0.4 mg doses.

GTN (Glyceryl Trinitrate) Spray

Pharmacokinetics

- GTN is rapidly absorbed from the oral mucosa and reaches the vascular system without passing through the liver.
- GTN is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects may be increased if the patient is taking an antihypertensive medicine.
- Severe and/or prolonged hypotension may occur if a phosphodiesterase inhibitor has been taken within the last 24 hours:
  - Phosphodiesterase inhibitors can be used in the treatment of erectile dysfunction and pulmonary hypertension (particularly sildenafil).
  - These medicines are long-acting vasodilators and the administration of GTN may cause further vasodilation.

Additional information

- GTN must be used with caution in the presence of STEMI because the risks may outweigh the benefits:
  - GTN may cause a significant fall in cardiac output.
  - GTN has a role in treating symptomatic myocardial ischaemia, but does not usually have a significant role in treating STEMI.
- Particular caution must be used in the presence of STEMI involving the right ventricle and personnel must have a low threshold for withholding GTN:
  - STEMI involving the right ventricle can result in a significant reduction in right ventricular contractility.
  - When the right ventricle is significantly impaired, it may provide little in the way of contribution to cardiac output and blood may be passively flowing down a pressure gradient between the inferior vena cava (IVC), the superior vena cava (SVC) and the left atrium.
  - This may result in the preload (filling) of the left side of the heart being dependent on the venous pressures within the IVC and SVC.
  - GTN can result in a significant fall in venous pressure (and thus a fall in preload) which may cause a significant fall in cardiac output.
- GTN and VT:

GTN (Glyceryl Trinitrate) Spray

- GTN is not administered for chest pain associated with VT because the risk of precipitating cardiac arrest outweighs the benefits.
- During VT the atria and ventricles are contracting independently of each other (also called atrioventricular dissociation). This causes a reduction in ventricular preload (or filling) through loss of the 'atrial kick', which in combination with abnormal ventricular contraction may cause a substantial fall in cardiac output.
- GTN reduces venous return to the heart and may further reduce cardiac output with the risk of precipitating severe hypotension or cardiac arrest.

More Information

Version history

16 June 2026 1.1.0.1
Update: Dosage updated.

11 September 2023 1.0.4
Update: Multiple changes.

22 November 2022 1.0.1
Update: Cautions.

Ropivacaine 0.75%

Mechanism of action

- Ropivacaine is a local anaesthetic. It blocks the initiation and transmission of nerve impulses by blocking the movement of sodium ions across the nerve cell membrane.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Severe pain associated with clinically obvious fractured neck of femur or fractured proximal shaft of femur, where pain has not been adequately controlled with an opiate and transport time (including extrication time) to hospital is greater than 30 minutes.

Contraindications

- ✕ Known severe allergy.
- ✕ Local infection at the site of injection.
- ✕ Additionally, for fascia iliaca block:
  - » Previous surgery in the groin.
  - » Aged less than 12 years.

Cautions

- ⚠ Taking an anticoagulant.

Use in pregnancy or when breastfeeding

- Safe and may be administered if indicated.

Dosage

- 40 ml of 0.375% (20 ml of 0.75% diluted to 40 ml) ropivacaine for an adult weighing greater than 60 kg.
- 30 ml of 0.375% (15 ml of 0.75% diluted to 30 ml) ropivacaine for an adult weighing 60 kg or less.

Ropivacaine 0.75%

- Administer the fascia iliaca block once only.

Administration

- Administer the dose described into the fascia iliaca space.

Common adverse effects

- Stinging at the time of injection.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 1-2 hours.

Usual preparation

- Ampoule containing 150 mg in 20 ml (0.75%).

Pharmacokinetics

- Ropivacaine is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.

Additional information

- A fascia iliaca block may be performed if the patient is taking antiplatelet medicines such as aspirin, clopidogrel or ticagrelor.

Ropivacaine 0.75%

- A fascia iliaca block may be performed if the patient is taking an anticoagulant but have a low threshold for stopping the procedure if it is difficult, and do not attempt multiple passes with the needle.
- Overdose of ropivacaine is very rare but can occur if doses exceed 150 mg in an adult or it is inadvertently administered IV. If this occurs the following may develop:
  - Tingling around the mouth.
  - Seizures.
  - Dysrhythmias, particularly bradydysrhythmias.
  - Hypotension.
  - Cardiac arrest.

More Information

Version history

22 November 2022 1.0.1

Update: Aligned with analgesia CPG.

Bleeding following Trauma

This guideline is for patients with suspected or confirmed internal bleeding, or clinically significant external bleeding, following blunt or penetrating trauma.

- Immediately control life-threatening external bleeding:
  - Apply manual external pressure.
  - Apply an arterial tourniquet if life-threatening bleeding from a limb is not easily controlled by manual external pressure.
  - Consider packing the wound if arterial tourniquet application is not indicated, using a haemostatic dressing if one is available.
- Apply a chest seal device to penetrating chest injuries or a standard dressing if a chest seal device is not available.
- Apply a pelvic splint if the mechanism of injury could have caused a fractured pelvis.
- Administer 0.9% sodium chloride IV (using a fluid warming device if available), if the patient has signs of hypovolaemia:
  - 500 ml IV for an adult.
  - 10 ml/kg IV for a child.
  - Repeat as required, minimising the total volume.
- Administer tranexamic acid IV:
  - 2 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- If shock is severe:
  - Expedite transport.
  - Call for blood, provided there is an established protocol for this in the area, particularly if extrication will be delayed and/or transport will be prolonged.
  - Consider seeking clinical advice and commencing a metaraminol infusion in an adult, if transport time is prolonged and the systolic BP is less than 80 mmHg systolic. Titrate the infusion rate to maintain a systolic BP of 80-90 mmHg.
- Provide active external warming whenever feasible.
- Splint long bone fractures.
- Notify receiving hospital personnel early. Activate a critical haemorrhage code if indicated, provided there is an established protocol for activation by ambulance personnel in the area.

Bleeding following Trauma

Backup

- Backup from a Paramedic must be requested if the patient has signs of hypovolaemia.
- Backup from an ICP/CCP must be requested if the patient has severe shock.
- Backup must not delay rapid transport to an appropriate hospital if shock is severe.

Referral and transport

- Expedite transport, particularly when there are signs of severe shock. This is particularly important with penetrating truncal trauma when the transport time to an appropriate hospital is short.
- Transport the patient direct to a major trauma hospital whenever this is feasible and safe.
- Transport the patient direct to a major trauma hospital with vascular surgical facilities if there are clear signs of arterial bleeding from a large artery, for example the brachial, popliteal or femoral artery. Clinical judgement must be used if a major trauma hospital with vascular surgical facilities is significantly further away than the closest major trauma hospital, and personnel should have a low threshold for seeking clinical advice in this setting.

Additional information

Arterial tourniquet application

- Applying a tourniquet is a balance of risk.
  - Tourniquets help control severe bleeding but can cause significant tissue ischaemia and damage (for example to nerves) and should only be used when external pressure is insufficient to control life-threatening bleeding.
  - Applying a tourniquet will cause ischaemia of the limb distal to the tourniquet and this may compromise the success of surgical reattachment.
- When applying a tourniquet:
  - Remove clothing from the limb if possible.
  - Apply as distally as possible.
  - Do not apply over a joint.
  - Tighten until bleeding clearly stops. The tourniquet must be tight enough to stop arterial flow.
  - Leave the wound exposed so that it can be observed for bleeding.
  - Record the time of application.

Bleeding following Trauma

- Reassess the tourniquet frequently following treatment. It may need to be further tightened if blood pressure improves.
- If the tourniquet is applied to a forearm or lower leg, the presence of two bones may limit the pressure that can be applied to vessels. If bleeding continues despite the tourniquet being tightened maximally, place the tourniquet on the upper arm or thigh.
- The tourniquet may need to be very tight if applied over a large thigh. In this setting a second tourniquet may be required proximal to the first tourniquet.
- Re-evaluate the need for a tourniquet after bleeding is controlled, IV access has been obtained and appropriate fluid resuscitation has been commenced:
  - If the source of the bleeding is a single lacerated artery, for example a lacerated brachial or radial artery, it is preferable that bleeding is controlled with direct pressure and then the tourniquet is released, particularly when transport time to hospital is over 30 minutes.
  - If bleeding cannot be controlled with direct pressure the tourniquet should be reapplied.
- A tourniquet may be used to provide pressure over a focal dressing, for example directly over a lacerated brachial artery. In this setting the tourniquet needs to be tight enough to control bleeding and not necessarily tight enough to stop arterial flow.

Pelvic splinting

- It is often difficult to determine accurately that a pelvis is fractured by clinical examination. Assume that the pelvis is fractured if the patient has pain in the pelvic area or is unable to report pain and has a suitable mechanism of injury.
- Remove clothing prior to placing a splint whenever feasible.
- Firmly splint the pelvis using a service supplied dedicated pelvic splint.
- Do not use a sheet to splint the pelvis because a sheet cannot be applied firmly enough.

Penetrating chest wounds and chest seal devices

- Use a service supplied dedicated chest seal provided one is available.
- Do not seal the wound with other devices (such as a defibrillator pad) as this risks the development of tension pneumothorax.
- A chest seal device is not required if the patient has been intubated and ventilated, as positive pressure in the chest from ventilation will prevent air entry into the chest via the wound.

Bleeding following Trauma

- Do not attempt to create a sealed dressing on three sides to form a valve, as this is rarely effective.

Administration of 0.9% sodium chloride

- 0.9% sodium chloride IV should be titrated to clinical signs of intravascular volume and perfusion.
- Large volumes of crystalloid may worsen bleeding by contributing to coagulopathy, particularly in the context of severe and uncontrolled bleeding:
  - This is why the total volume of 0.9% sodium chloride should be minimised.
  - Clinical judgment is required when determining the volume administered, but it should be uncommon to exceed one litre in an adult or 20 ml/kg in a child, and it should be rare to exceed two litres in an adult or 40 ml/kg in a child.
  - Have a low threshold for seeking clinical advice if shock is severe and transport time to the most appropriate hospital is prolonged.
- Ambient temperature IV fluid can cause hypothermia which worsens bleeding by contributing to coagulopathy. IV fluid should be warmed using a dedicated warming device if one is available.
- Do not warm IV fluid in microwaves because this can result in unreliable fluid temperatures and/or may damage the bag.

Tranexamic acid (TXA)

- TXA reduces bleeding by reducing clot break down via fibrinolysis, and this is associated with reduced mortality rates in bleeding secondary to trauma.
- TXA is indicated if there are signs of clinically significant bleeding or signs of hypovolaemia.
- TXA administration is not a priority and it should usually be administered with the first dose of 0.9% sodium chloride.

Blood

- Requests for blood to be delivered to the scene must not compromise rapid transport to the most appropriate hospital.
- Blood should only be requested if:
  - Shock is severe, and
  - There is an established protocol in the area for blood to be delivered to the scene, and

Bleeding following Trauma

- Blood can be administered 15 or more minutes before the patient can arrive in the most appropriate hospital.

- Blood may be available via staging at a small hospital/medical facility en route to the most appropriate hospital. In this setting personnel must have a low threshold for seeking advice from the on-call doctor for the service.

- ICPs/CCPs may continue blood administration initiated by a medical practitioner following their verbal instructions but should have a low threshold for seeking clinical advice.

- Requests for blood to be delivered to the scene outside of an established protocol must be made via the on-call doctor for the service. In this setting:
  - It will usually take approximately 15 minutes to arrange for blood to be prepared, plus transport time to the scene. This means that for such a request to be appropriate, the patient should usually have severe shock and a prolonged transport time.
  - Personnel cannot administer blood without specific instructions from the on-call doctor for the service.

Patient temperature

- It is important to keep the patient warm because hypothermia worsens bleeding by contributing to coagulopathy.

- Passive warming (for example with external blankets) is rarely effective, and the patient should also be actively warmed.

- It is important to document the patient's temperature.

Bleeding from wounds

- The best method of controlling severe external bleeding is usually by providing firm and sustained manual pressure using a dressing directly over the site of the bleeding, before applying a focal dressing and a pressure bandage.

- A haemostatic dressing (if available) will reduce bleeding, but do not delay manual external pressure while obtaining a haemostatic dressing.

- Topical adrenaline is not useful in the setting of severe external bleeding. Topical adrenaline may be considered in the setting of non-severe bleeding if a haemostatic dressing is unavailable.
  - Dilute adrenaline to 1:10,000 using 0.9% of sodium chloride. For example, dilute 2 mg of adrenaline to a total of 20 ml.
  - Soak a dressing in 1:10,000 adrenaline. Place the dressing into the wound or over the bleeding site, and continue to provide pressure over the wound.

Bleeding following Trauma

Critical haemorrhage codes and the assessment of blood consumption score

- Critical haemorrhage codes such as Code Crimson and Code Red are used in many major trauma hospitals to accelerate in-hospital clinical interventions to stop life-threatening bleeding.
- Notify ED personnel as soon as possible if shock is severe, to enable early activation of a critical haemorrhage code. Where an agreed protocol is in place this may be activated by transporting personnel.
- The criteria for critical haemorrhage code activation are:
  - An assessment of blood consumption (ABC) score greater than or equal to 2, or
  - The patient has received blood products.
- The ABC score is composed of (score one for each):
  - Heart rate greater than or equal to 120/min.
  - Systolic BP less than or equal to 90 mmHg.
  - Penetrating truncal trauma.
  - A positive ultrasound scan for free fluid in the abdomen, pelvis, or chest.

Penetrating objects

- Penetrating objects should usually be removed from a limb, as this will allow direct pressure to be more reliably applied. Clinical judgement should be used if the object is large (for example a knife rather than a nail) and appears to be near an artery.
- Do not remove penetrating objects from the head or truncal area because severe bleeding may occur that cannot be compressed.
- Provide manual pressure directly around the object if severe bleeding is present and the object is not removed.

Metaraminol

- Metaraminol is not a priority in patients with haemorrhage. However, as shock progresses vasodilation will often develop, and it is important to minimise the total volume of IV fluid administered.
- Metaraminol should only be administered to adult patients who continue to have severe hypotension following IV fluid administration.
- If a metaraminol infusion has been commenced, this should be accompanied by further fluid boluses if indicated.

Bleeding following Trauma

- The aim of administering metaraminol is not to restore normal perfusion, but to maintain sufficient perfusion of vital organs without increasing blood loss.

More Information

Version history

28 October 2024 1.0.5

New: Guideline released.

Lignocaine 1%

Mechanism of action

- Lignocaine is a local anaesthetic.
- Lignocaine blocks the initiation and transmission of nerve impulses by blocking the movement of sodium ions across the nerve cell membrane.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Subcutaneous injection for prophylaxis of pain associated with IV cannulation.
- ✓ Subcutaneous injection for digital ring blocks for analgesia.
- ✓ Intraosseous injection for significant bone pain associated with intraosseous administration.

Contraindications

- ✓ Known severe allergy.
- ✓ Local infection in the area of injection.

Cautions

- ⚠ Taking an anticoagulant (ring blocks).

Use in pregnancy or when breastfeeding

- Safe. May be administered if indicated.

Dosage

- Subcutaneous:
  - The maximum subcutaneous dose for an adult is 200 mg (20 ml of 1% lignocaine).
  - See the paediatric drug dose tables for the maximum dose for a child.

Lignocaine 1%

- The subcutaneous dose may be repeated once after 30 minutes.

- Intraosseous:
  - 50 mg (5 ml of 1% lignocaine) for an adult.
  - See the paediatric drug dose tables for a child.
  - The intraosseous dose may be repeated once after 15 minutes.

Administration

- Subcutaneous for IV insertion: administer into the subcutaneous tissue at the site of cannulation. Raise a bleb and wait approximately one minute before insertion.
- Digital ring blocks: administer approximately 1-2 ml of 1% lignocaine into the tissue on either side of the web space of the digit.
- Intraosseous: administer slowly over 1-2 minutes and wait one further minute before infusing fluid. This is intended to limit the amount of lignocaine flushed into the circulation.

Common adverse effects

- Stinging at the time of injection.

Usual onset of effect

- 1-2 minutes for IV cannulation.
- 5-10 minutes for digital ring blocks.

Usual duration of effect

- 30-60 minutes.

Usual preparation

- Ampoule containing 50 mg in 5 ml.

Lignocaine 1%

Pharmacokinetics

- Lignocaine is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.

Additional information

- Do not apply lignocaine topically to the eye because the solution contains a preservative that may cause harm.
- Warming lignocaine, for example in your pocket or hand, may reduce the stinging associated with subcutaneous injection.
- Overdose of lignocaine when administered subcutaneously is very rare, but can occur if doses exceed 3 mg/kg or more than 1 mg/kg is inadvertently administered IV. If this occurs the following may develop:
  - Tingling around the mouth.
  - Seizures.
  - Dysrhythmias, particularly bradydysrhythmias.
  - Hypotension.
  - Cardiac arrest.

More Information

Version history

16 June 2026 1.1.0.1
Update: Indications updated.

22 November 2022 1.0.1
Update: Aligned with analgesia CPG.

Hypoglycaemia

This guideline is for patients with a blood glucose concentration less than 3.5 mmol/litre. See the **'neonatal resuscitation'** guideline if the patient is a newborn baby.

- Administer 10-20 g of glucose PO provided the patient is sufficiently conscious to swallow safely.
- Administer glucose IV if the patient has a significantly altered level of consciousness or cannot swallow safely:
  - 100 ml of 10% glucose IV for an adult.
  - 2 ml/kg of 10% glucose IV for a child.
- Administer glucagon if unable to gain IV access:
  - 1 mg of glucagon IM for a patient aged five years and over.
  - 0.5 mg of glucagon IM for a child aged less than five years.
- Repeat the glucose measurement every ten minutes until the glucose concentration is consistently greater than 3.5 mmol/litre. Administer further doses of glucose if required, but do not repeat glucagon IM.

Referral and transport

- A patient may receive treatment for hypoglycaemia and be given a clear recommendation for treatment without transport providing the following criteria are met:
  - No red flags are present.
  - A competent adult can stay with the patient for the next four hours.
  - The patient eats a meal (preferably with complex carbohydrates).
  - The patient is given the hypoglycaemia information sheet, which is explained to them and the accompanying adult.
  - The patient is given a clear recommendation to have their treatment reviewed (for example by an appropriate healthcare professional or diabetes service personnel). If the patient is aged less than or equal to 18 years or has been recently diagnosed with diabetes, this review should occur within 24 hours if feasible.

Red flags (Should be seen in ED, usually via ambulance)

Hypoglycaemia

- Overdose (including accidental) of insulin or oral hypoglycaemics.
- No known history of diabetes.
- Non-isolated episode.
- Taking an oral hypoglycaemic medicine and has hypoglycaemia without an obvious cause.
- Does not recover fully.
- Unable to mobilise normally.
- Unable to achieve a blood glucose concentration greater than 3.5 mmol/litre, ten minutes (or more) after glucagon or the last glucose administration.

Additional information

Causes of hypoglycaemia

- Hypoglycaemia usually occurs in a patient taking insulin or an oral hypoglycaemic medicine.
- Less common causes of hypoglycaemia include severe sepsis (particularly in young children), poisoning with medicines that lower blood glucose and liver failure.
- Some oral hypoglycaemic medicines are excreted primarily by the kidneys. If a patient taking an oral hypoglycaemic medicine develops hypoglycaemia without an obvious cause, unrecognised deterioration in kidney function may be a contributor.

Measuring blood glucose concentration

- Do not measure blood glucose concentration using the patient's glucose meter because the result may be inaccurate.
- Use capillary blood and not blood from other sources because glucose meters are calibrated to use capillary blood.
- Check the history of the patient's glucose meter whenever possible. In particular rule out recurrent hypoglycaemia.

Treating hypoglycaemia with oral glucose

- Oral glucose is best administered as a simple carbohydrate that is rapidly absorbed.

Hypoglycaemia

- Examples include 10% glucose (100 ml contains 10 g of glucose), glucose tablets or gel, sugar dissolved in water, 'non-diet' jam or similar glucose containing spreads and 'non-diet' soft drinks.
- 10-20 g of glucose is approximately half a cup of 'non-diet' fruit juice or soft drink, one tablespoon of sugar, one tablespoon of honey or six squares of chocolate.
- Most oral glucose gels contain 10-20 g of glucose per dose and can be administered to all patients, regardless of age.
- Document oral glucose as the approximate number of grams of oral glucose administered, or a description of the food ingested.

- Glucose gel may be spread on the gums, tongue and inside of the cheeks of a baby or small child.
- Hypoglycaemia may occur several hours later as glucose is metabolised. This is why the instructions within the information sheet are important and why a meal must be eaten. The meal should preferably contain a complex carbohydrate such as brown rice, whole grain bread, porridge or beans.

Treating hypoglycaemia with glucagon

- Glucagon will be ineffective if the patient has depleted glycogen stores. Examples include if the patient does not have diabetes, is a young child, has sepsis, or has not eaten food for more than 12 hours. In this setting glucagon will not be harmful, but the focus must be on administration of glucose.

Hypoglycaemia in children aged less than five years

- Hypoglycaemia in children aged less than five years is most commonly due to:
  - An illness in combination with no food intake and depleted glycogen stores, or
  - Severe sepsis, or
  - Poisoning with oral hypoglycaemic medicines.
- Children not known to have diabetes are usually malnourished or glycogen depleted for another reason, and glucagon is unlikely to be effective.
- Although metabolic disorders can cause hypoglycaemia in young children, this is rare. Do not delay treatment of hypoglycaemia in young children pending diagnostic tests, other than measuring a capillary glucose concentration. Although some references advocate delaying treatment until a blood sample is taken for analysis of possible metabolic disorders, treatment always takes priority over blood tests in this setting.

Response and/or transport by helicopter

Hypoglycaemia

- Response by helicopter may be indicated if it will arrive at scene at least 30 minutes sooner than a road-based Paramedic, and the patient is unconscious.

More Information

Version history

11 September 2023 1.0.4
Update: Multiple changes.

Sodium Bicarbonate 8.4%

Mechanism of action

- 8.4% sodium bicarbonate is a systemic alkalinising agent. It increases plasma bicarbonate, buffers hydrogen ions and raises the blood pH.
- Following crush injury 8.4% sodium bicarbonate has a role because:
  - Sodium ions help protect cardiac cell membranes from the effects of hyperkalaemia.
  - A rise in pH results in potassium moving into the cells, which lowers the potassium concentration in blood.
  - A rise in urinary pH reduces myoglobin deposition in the kidneys.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Release syndrome following crush injury in an adult.
- ✓ Known or suspected hyperkalaemia with severe ECG changes.
- ✓ Cardiac arrest secondary to hyperkalaemia.
- ✓ Suspected cyclic antidepressant poisoning with QRS prolongation.
- ✓ Severe TBI and intubated with clinical signs of raised intracranial pressure.

Contraindications

- ✗ Known severe allergy.

Cautions

- ⚠ IV access via a small vein. 8.4% sodium bicarbonate is hyperosmolar and will cause venous irritation if administered via a small vein.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but 8.4% sodium bicarbonate should be administered if indicated.

Sodium Bicarbonate 8.4%

Dosage

- Cyclic antidepressant poisoning:
  - 100 ml IV for an adult.
  - 2 ml/kg for a child.
- Severe TBI, intubated and clinical signs of raised intracranial pressure:
  - 100 ml IV for an adult.
  - 2 ml/kg for a child.
- Hyperkalaemia with severe ECG changes:
  - 100 ml IV for an adult.
  - Repeat the dose after 20 minutes if severe ECG changes persist or recur.
  - Seek clinical advice for a child.
- Cardiac arrest secondary to hyperkalaemia:
  - 100 ml IV for an adult.
  - Repeat the dose if ROSC is not achieved within 10 minutes of the first dose.
  - Seek clinical advice for a child.
- Release syndrome following crush injury:
  - 100 ml IV for an adult.
  - Administer further doses if signs of hyperkalaemia (including dysrhythmia) occur.
  - Seek clinical advice for a child.

Administration

- Cardiac arrest secondary to hyperkalaemia: Administer IV as a bolus, via a running line.
- All other indications: Administer IV over approximately one minute via a running line.
- Do not mix with other medicines as precipitation will occur. If other medicines are being administered via the same vein, ensure a minimum IV flush of 50 ml of 0.9% sodium chloride between medicines.

Common adverse effects

- None.

Sodium Bicarbonate 8.4%

Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 1-2 hours.

Usual preparation

- Ampoule containing 1 mmol/ml of sodium bicarbonate.
- Ampoules may be 10 ml, 50 ml or 100 ml.

Pharmacokinetics

- 8.4% sodium bicarbonate dissociates into sodium and bicarbonate ions which are excreted in urine.
- There are no significant effects from renal impairment on administration.

Common interactions

- None.

Additional information

- Document the dose administered as mmol.

More Information

Version history

28 October 2024 1.0.5
Update: Indications and dosage updated.

11 September 2023 1.0.4
Update: New indications have been added. Dosage and administration information updated.

Adenosine

Mechanism of action

- Adenosine is an antidysrhythmic used for the treatment of paroxysmal supraventricular tachycardia (SVT).
- Adenosine is a nucleoside that depresses conduction through the AV node. This interrupts re-entry circuits within the heart and may restore sinus rhythm in patients with SVT.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Patients aged greater than or equal to 12 years with SVT and a ventricular rate greater than or equal to 150/minute, and
  - The patient is not severely compromised, and
  - The rhythm fails to revert following two Valsalva manoeuvres.

Contraindications

- ✕ Known severe allergy.
- ✕ Known sick sinus syndrome without an internal pacemaker in place. Adenosine may cause severe bradycardia if the patient has sick sinus syndrome.
- ✕ Previous 2nd or 3rd degree heart block without an internal pacemaker in place. Adenosine may cause heart block if the patient has had previous heart block.
- ✕ Previous heart transplantation without an internal pacemaker in place. Following a heart transplant the heart is denervated and adenosine may cause severe bradycardia.

Adenosine

Cautions

- Asthma. Adenosine may precipitate bronchospasm and should be withheld if the patient has had recurrent life-threatening attacks of bronchospasm, or is currently suffering an exacerbation of asthma.
- COPD. Adenosine may precipitate bronchospasm and should be withheld if the patient has had recurrent life-threatening attacks of bronchospasm, or is currently suffering an exacerbation of COPD.
- Wolff-Parkinson-White (WPW) syndrome if the rhythm is possibly fast atrial fibrillation. Adenosine is not contraindicated in a patient with known WPW syndrome provided the rhythm is clearly SVT. If the rhythm is possibly fast atrial fibrillation, adenosine should be withheld because of the risk of precipitating VF.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. However, the balance of risk is in favour of administration if indicated.

Dosage

- 6 mg.
- A second dose of 12 mg may be administered if the rhythm fails to revert.

Administration

- The preferred site of injection is the antecubital fossa.
- Administer the dose as a rapid bolus followed immediately by a rapid flush of a minimum of 20 ml of 0.9% sodium chloride.

Common adverse effects

- Bradycardia and/or sinus pause which may be up to 30 seconds.
- Shortness of breath and/or an urge to breathe deeply.
- Feeling light-headed.
- Nausea and flushing.
- Feeling of chest pressure and/or severe apprehension.

Adenosine

Usual onset of effect

- 5-10 seconds.

Usual duration of effect

- 10-20 seconds.

Pharmacokinetics

- Adenosine is rapidly taken up and metabolised within seconds by red blood cells and vascular endothelial cells.

Usual preparation

- Ampoule containing 6 mg in 2 ml.

Common interactions

- None.

Additional information

- Adenosine usually causes a brief period of very low cardiac output and this often causes the patient to feel severe apprehension and/or an impending sense of doom. Warn the patient they may feel awful but reassure them this will pass very quickly.
- If the rhythm is possibly fast atrial fibrillation and an accessory pathway is present, adenosine may precipitate VF. If uncertain, treat the rhythm as atrial fibrillation and not SVT.
- Seek clinical advice if the patient is aged less than 12 years.

More Information

Version history

11 September 2023 1.0.4

Update: Indications updated. Addition to administration and additional information headings.

Adenosine

Adrenaline

Mechanism of action

- Adrenaline stimulates alpha and beta receptors, with the predominant effects occurring at alpha 1, beta 1 and beta 2 receptors.
- Alpha 1 stimulation causes smooth muscle contraction, vasoconstriction of blood vessels and stimulation of glycogenolysis and gluconeogenesis.
- Beta 1 stimulation causes an increase in inotropy (cardiac contractility), an increase in chronotropy (heart rate) and an increase in dromotropy (speed of electrical conduction within the heart).
- Beta 2 stimulation causes smooth muscle relaxation, skeletal muscle vasodilation, bronchodilation, and stabilisation of mast cell membranes, reducing histamine release.

Authorised practice levels

- EMTs: nebulised, IM, IN and topical adrenaline.
- Paramedics: all of the above and adrenaline IV for cardiac arrest.
- ICPs and CCPs: all indications and all routes.

Indications

- Cardiac arrest.
- Anaphylaxis.
- Severe asthma.
- Imminent respiratory arrest from COPD.
- Bradycardia with moderate or severe cardiovascular compromise.
- Blood pressure support if unresponsive to metaraminol.
- Septic shock, cardiogenic shock, and neurogenic shock, unresponsive to 0.9% sodium chloride IV and metaraminol IV.
- Stridor causing moderate or severe respiratory distress.
- IN for clinically significant epistaxis.
- Topical for clinically significant bleeding from a wound.

Adrenaline

Contraindications

- None.

Cautions

- Myocardial ischaemia. Adrenaline will increase myocardial oxygen consumption.
- Tachydysrhythmias. Adrenaline will usually make tachydysrhythmias worse.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

Dosage

- The dose of adrenaline is dependent on the indication and the route. See the individual guideline.

Administration

- Topical: dilute each mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Apply topically in addition to direct pressure.
- IN: dilute each mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Administer the appropriate dose into each bleeding nostril using a mucosal atomising device, in addition to direct pressure.
- Nebulised: administer undiluted.
- IM: administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.
- Cardiac arrest:
  - Adults and children whose weight has been rounded to 50 kg or more: administer undiluted as an IV bolus.
  - Children whose weight has been rounded to 40 kg or less: dilute 1 mg of adrenaline to a total of 10 ml using 0.9% sodium chloride. This solution is 1:10,000 and contains 0.1 mg/ml. Draw up the dose from this solution and administer as an IV bolus.

Adrenaline

- IV infusion: place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride. Shake well and label. This solution is 1:1,000,000 and contains 0.001 mg/ml.
  - For an adult: administer as an IV infusion starting at 2 drops/second. Adjust the rate to the patient's condition.
  - For a child aged 5-14 years: administer as an IV infusion starting at 1 drop/second. Adjust the rate to the patient's condition.
- When administering an IV infusion of adrenaline using 1 mg of adrenaline in a 1 litre bag of 0.9% sodium chloride:
  - 2 drops/second via a standard IV administration set will administer approximately 0.4 mg/hour of adrenaline.
  - Record an estimate of the total dose of adrenaline administered.
- For all other IV administration: place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride, Shake well and label. This solution is 1:1,000,000 and contains 0.001 mg/ml. Draw up the dose from this solution and administer as an IV bolus.

Common adverse effects

- Tachycardia.
- Tachydysrhythmia.
- Myocardial ischaemia.
- Ventricular ectopy.
- Hypertension.
- Nausea and vomiting.
- Tremor, anxiety and sweating.
- Hyperglycaemia.

Usual onset of effect

- IV: 5-10 seconds.
- IM: 2-5 minutes.
- Nebulised, IN and topical: on contact with the target site.

Adrenaline

Usual duration of effect

- The cardiovascular effects last 5-15 minutes.
- The mast cell membrane effects may last for several hours.

Usual preparation

- Ampoule containing 1 mg in 1 ml.

Pharmacokinetics

- Adrenaline is metabolised by the liver and taken up by sympathetic nerve endings.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Increased doses may be required if the patient is taking a beta-blocker or a calcium channel blocker. This effect is particularly prominent in the setting of poisoning if a large dose of a beta-blocker and/or calcium channel blocker has been taken.

More Information

Version history

22 November 2022 1.0.1
Update: Indication to align with stridor and croup CPGs.

Amiodarone

Mechanism of action

- Amiodarone is an antidysrhythmic with a broad spectrum of activity.
- Amiodarone has predominantly class III activity. It prolongs the action potential duration, reduces automaticity and prolongs the refractory period of atrial, nodal and ventricular tissues.
- The electrophysiological effects result in a reduction in abnormal electrical activity (for example ectopy), a reduction in electrical conduction, a reduction in heart rate and a stabilisation of the SA and AV nodes.
- Amiodarone also causes a small increase in coronary blood flow (although this is not usually clinically significant) and a reduction in myocardial oxygen consumption by reducing inotropy (the force of cardiac contraction).

Authorised practice levels

- Paramedics: cardiac arrest.
- ICPs and CCPs: all indications.

Indications

- ✓ Cardiac arrest with VF or VT after three shocks.
- ✓ Adults with sustained VT in the absence of cardiac arrest.
- ✓ Adults with moderate cardiovascular compromise as a result of atrial fibrillation, where the ventricular rate is predominantly greater than 130/minute and metoprolol PO is contraindicated.
- ✓ Adults with moderate cardiovascular compromise as a result of atrial flutter, where the ventricular rate is predominantly greater than 130/minute.

Contraindications

- ✗ Known severe allergy.
- ✗ Known severe allergy to iodine.
- ✗ VT secondary to cyclic antidepressant poisoning. In this setting amiodarone administration can be associated with severe worsening of shock, without resolution of the rhythm.
- ✗ If torsades de pointes is present or has occurred.

Amiodarone

Cautions

- None if the patient is in cardiac arrest (with the exception of arrests involving tricyclic antidepressant poisoning or Torsades de Pointes, see contraindications).
- Poor perfusion or signs of low cardiac output. Amiodarone reduces inotropy and may cause a significant fall in cardiac output, particularly when administered rapidly.
- Hypotension. Amiodarone causes vasodilation and may worsen hypotension, particularly when administered rapidly.
- Atrial fibrillation associated with sepsis. Amiodarone may cause a significant fall in cardiac output.
- Known sick sinus syndrome without an internal pacemaker in place. Amiodarone slows the heart rate and severe bradycardia may occur following reversion of a tachydysrhythmia.
- Previous 2nd or 3rd degree heart block without an internal pacemaker in place. Amiodarone slows the heart rate and severe bradycardia may occur following reversion of a tachydysrhythmia.
- Pregnancy.

Use in pregnancy or when breastfeeding

- May cause harm during pregnancy. Do not administer amiodarone unless there is a strong clinical indication.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- Cardiac arrest:
  - 300 mg for an adult.
  - If in VF or VT after six shocks, a second dose of 150 mg may be administered.
  - See the paediatric drug dose tables for a child.
- Tachydysrhythmia in an adult:
  - 300 mg IV over 30 minutes.
  - A further 150 mg IV over 30 minutes may be administered if the tachydysrhythmia persists.

Amiodarone

Administration

- Cardiac arrest: administer IV undiluted as a bolus.
- Tachydysrhythmia:
  - Place 300 mg of amiodarone in 100 ml of 5% glucose and label. 1 drop/second via a standard IV administration set will deliver 100 ml over approximately 30 minutes. Slow the rate of infusion if hypotension occurs.
  - The administration set will need to be flushed with 0.9% sodium chloride to ensure that all of the amiodarone has been administered.
  - An IV infusion over 30 minutes is the preferred method of administration. However, it is acceptable to dilute 300 mg of amiodarone to a total volume of 20-30 ml using 5% glucose or 0.9% sodium chloride. Administer this IV over 30 minutes and slow the rate of infusion if hypotension occurs.

Common adverse effects

- Hypotension.
- Lightheadedness.
- Bradydysrhythmia.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 1-4 hours after a single dose.
- Amiodarone is taken up into tissues and slowly released. This may result in a prolonged half-life, particularly when more than one dose has been administered. This is why many references quote a half-life of 10-60 days, but the clinical duration of effect is much shorter than this.

Usual preparation

- Ampoule containing 150 mg in 3 ml.

Amiodarone

Pharmacokinetics

- Amiodarone is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- May potentiate the action of cyclic antidepressants in cyclic poisoning.
- May cause bradycardia following reversion of dysrhythmia if the patient is taking a beta-blocker and/or a centrally acting calcium channel blocker (for example diltiazem).

Additional information

- If the indication is atrial fibrillation causing moderate cardiovascular compromise, the goal of treatment is to control the ventricular rate and not to revert the rhythm to sinus rhythm, although treatment with amiodarone may result in reversion of the rhythm to sinus rhythm. If a patient has been in atrial fibrillation for longer than a few days, there is a small risk that this may be associated with emboli leaving the left atrium. This is why amiodarone is reserved for patients with cardiovascular compromise that is clinically significant.
- If amiodarone is commenced, the full dose should be administered even if the rhythm reverts to sinus rhythm, unless severe hypotension or bradycardia occurs.
- Amiodarone is often described as relatively contraindicated in the presence of a prolonged QT interval, but this only applies to long-term administration.
- When diluting amiodarone with 0.9% sodium chloride the solution will go slightly cloudy. This does not adversely affect the amiodarone.
- Avoid shaking the ampoule because this will cause the solution to froth.

More Information

Version history

16 June 2026 1.1.0.1

Update: Indications, contraindications and dosage updated.

Atropine

- Atropine is an anticholinergic which is mostly used for the treatment of bradycardia.
- Atropine antagonises (blocks) muscarinic acetylcholine receptors, causing vagal inhibition resulting in:
  - An increase in heart rate.
  - Drying of salivary and bronchial secretions.
  - Bronchodilation.
  - Reduced gastrointestinal motility.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Adults with bradycardia, particularly if the rhythm is narrow complex.
- ✓ Organophosphate poisoning.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Myocardial ischaemia. Atropine will increase myocardial oxygen consumption.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 0.6 mg for an adult. Repeat as required without a maximum dose, if the bradycardia is responsive to atropine.

Atropine

- Repeated and escalating doses are likely to be required for organophosphate poisoning.

Administration

- Administer undiluted as a rapid IV bolus. Slow administration may result in transient bradycardia.

Common adverse effects

- Tachycardia.
- Confusion. Particularly in the elderly or those with intellectual impairment.
- Dry mouth.
- Blurred vision.

Usual onset of effect

- 5-10 seconds.

Usual duration of effect

- The cardiovascular effects last 15-60 minutes.
- The exocrine and smooth muscle effects last 4-6 hours.

Usual preparation

- Ampoule containing 0.6 mg in 1 ml.

Pharmacokinetics

- Atropine is predominantly metabolised in the liver, but some is excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

Atropine

- The action of atropine may be potentiated if the patient is taking other drugs with anticholinergic properties, such as phenothiazines, some antihistamines (such as promethazine, but not loratadine), tricyclic antidepressants and anti-parkinsonian medicines. These interactions are rarely clinically significant.

More Information

Version history

11 September 2023 1.0.4

Update: Indications updated to align with changes to bradycardia guideline.

Calcium Chloride

Mechanism of action

- Calcium is the active ingredient in calcium chloride.
- Calcium is a mineral that is essential for a number of normal body functions including cell membrane function, enzyme reactions, transmission of nerve impulses, cardiac electrophysiology, contraction of cardiac and skeletal muscle, and coagulation.
- Calcium raises the cardiac action potential threshold and protects cardiac cell membranes from the effects of hyperkalaemia, resulting in a reduction in dysrhythmias associated with hyperkalaemia.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Release syndrome following crush injury in adults.
- ✓ Suspected hyperkalaemia with severe ECG changes.
- ✓ Cardiac arrest secondary to suspected hyperkalaemia.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but calcium should be administered if indicated.

Dosage

- Crush injury:

Calcium Chloride

- 6.8 mmol (1 g) of calcium chloride IV over one minute for an adult.
- Repeat the dose if signs of hyperkalaemia persist or recur.
- Seek clinical advice for a child.

- Hyperkalaemia with severe ECG changes:
  - 6.8 mmol (1 g) of calcium chloride IV over one minute for an adult.
  - Repeat the dose after 20 minutes if signs of severe ECG changes persist or recur.
  - Seek clinical advice for a child.

- Cardiac arrest secondary to hyperkalaemia:
  - 6.8 mmol (1 g) of calcium chloride IV as a bolus for an adult.
  - Repeat the dose after 10 minutes if ROSC is not achieved.
  - Seek clinical advice for a child.

Administration

- Crush injury and hyperkalaemia with severe ECG changes:
  - Administer IV over 1 minute into a large vein via a running line, as this reduces venous irritation.

- Cardiac arrest secondary to hyperkalaemia:
  - Administer as a bolus via a running line.

- Do not mix with other medicines (and in particular do not mix with sodium bicarbonate) as precipitation will occur. If other medicines are being administered via the same vein, ensure a minimum flush of 50 ml of 0.9% sodium chloride between medicines.

Common adverse effects

- Venous irritation including redness and pain at the site of injection.
- Tingling sensation.
- Rapid administration may cause dysrhythmias.

Usual onset of effect

- 2-5 minutes.

Calcium Chloride

Usual duration of effect

- 1-4 hours.

Usual preparation

- Ampoule containing 6.8 mmol (1 g) in 10 ml.

Pharmacokinetics

- 80% is excreted in faeces and 20% is excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

- None.

Additional information

- Document the dose administered in mmol.
- Calcium gluconate is an alternative formulation of calcium, usually containing 2.2 mmol/10 ml. Administer two ampoules at a time if only calcium gluconate is available.

More Information

Version history

11 September 2023 1.0.4

Update: Indications updated. Information under dosage and administration headings updated.

Cefazolin

Mechanism of action

- Cefazolin is a first-generation cephalosporin antibiotic with activity against gram-negative and gram-positive bacteria. It inhibits production of the bacterial cell wall, causing bacteria to die.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- Sepsis, where the source of infection appears to be the soft tissues or joint, and:
  - The patient is aged greater than or equal to 12 years, and
  - Has one or more clinical features indicating antibiotics, and
  - Time to hospital is greater than 30 minutes.
- Compound fracture
- Large contaminated wound.
- Following chest decompression.
- Following amputation.
- Cellulitis. In this setting a single dose may be administered if the patient is not being directly referred (or transported) to a medical facility.

Contraindications

- Known severe allergy to cephalosporins.

Cautions

- None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Cefazolin

Dosage

- 1 g IV for an adult.
- See the paediatric drug dose tables for a child.

Administration

- Add approximately 4 ml of 0.9% sodium chloride to a 1 g ampoule and shake until dissolved.
- Draw up the ampoule contents and dilute to a total of 10 ml.
- Administer IV over 1-2 minutes preferably into a running line.

Common adverse effects

- None.

Usual onset of effect

- 30-60 minutes

Usual duration of effect

- 6-8 hours.

Usual preparation

- Ampoule containing 1 g of cefazolin as a powder for reconstitution.

Pharmacokinetics

- Cefazolin is predominately excreted in the urine.
- Clearance is prolonged if the patient has significant kidney impairment, but this does not alter the initial (loading) dose.

Cefazolin

Common interactions

- None.

Additional information

- Rapid administration of cefazolin can cause pain and phlebitis, which is why it should be preferably administered into a running line.

More Information

Version history

28 October 2024 1.0.5
Update: Indications updated.

11 September 2023 1.0.4
New: Guideline released.

Ceftriaxone

Mechanism of action

- Ceftriaxone is a cephalosporin antibiotic with broad activity against gram-negative and gram-positive bacteria. It inhibits production of the bacterial cell wall, causing bacteria to die.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Suspected meningococcal septicaemia.
- ✓ Sepsis, where cefazolin is not indicated, and:
  - The patient is aged greater than or equal to 12 years, and
  - One or more clinical features indicating antibiotics are present, and
  - Time to hospital is greater than 30 minutes.

Contraindications

- ✕ Anaphylaxis to cephalosporins.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- 2 g IV for an adult.
- 2 g IM for an adult if IV access cannot be immediately obtained (meningococcal septicaemia only).
- See the paediatric drug dose tables for a child.

Ceftriaxone

Administration

- IV administration:
  - Add approximately 4 ml of 0.9% sodium chloride to a 2 g ampoule and shake until dissolved.
  - Draw up the ampoule and dilute to a total of 10 ml.
  - Discard unrequired volume before administration for a child.
  - Administer IV over 1-2 minutes preferably into a running line.
- IM administration:
  - Add 4 ml of 0.9% sodium chloride to a 2 g ampoule and shake until dissolved.
  - Draw up the ampoule using two syringes with approximately half in each. The total volume will be 5 ml.
  - Discard unrequired volume before administration for a child.
  - Administer one syringe into each lateral thigh. If this site is not suitable, use each lateral upper arm.

Common adverse effects

- None.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 24 hours.

Usual preparation

- Ampoule containing 2 g of ceftriaxone as a powder for reconstitution.

Pharmacokinetics

- 50% is excreted in urine and 50% in bile.
- Neither renal impairment nor hepatic impairment alter the initial (loading) dose.

Ceftriaxone

Common interactions

- None.

Additional information

- Some references describe dissolving ceftriaxone using 1% lignocaine for IM injection, to reduce the pain of injection. In the setting of suspected meningococcal septicaemia this reduction in pain is not significant and ceftriaxone should be dissolved using 0.9% sodium chloride.
- As meningococcal bacteria die they release endotoxins. The body's immune response to endotoxins can cause profound worsening of shock following antibiotic administration. Be prepared to treat this with 0.9% sodium chloride IV, and metaraminol or adrenaline IV. It is rare for significant amounts of endotoxins to be released from other bacteria.

More Information

Version history

11 September 2023 1.0.4
Update: Indications updated.

Clopidogrel

Mechanism of action

- Clopidogrel has antiplatelet activity.
- Clopidogrel antagonises (blocks) the binding of adenosine diphosphate (ADP) to platelets and impairs platelet function. Clopidogrel provides significantly more antiplatelet activity than aspirin.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ STEMI in conjunction with fibrinolytic therapy.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Clinically significant bleeding. Clopidogrel will increase bleeding.
- ⚠ At risk of bleeding. If there are any cautions present within the fibrinolytic therapy checklist, personnel must seek clinical advice.
- ⚠ Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. The likelihood of STEMI occurring in a woman who is pregnant is so low that personnel must seek clinical advice.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 300 mg if the patient is aged less than 75 years.

Clopidogrel

- 75 mg if the patient is aged greater than or equal to 75 years.

Administration

- Administer PO.

Common adverse effects

- Increased bleeding.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 3-5 days. This is because platelets exposed to clopidogrel are impaired for the life of the platelet which is 7-10 days. Approximately 10% of platelets are replaced each day.

Usual preparation

- 75 mg tablets.

Pharmacokinetics

- Clopidogrel is a prodrug and must be metabolised to the active form in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The risk of bleeding will be increased if the patient is taking an anticoagulant.

More Information

Clopidogrel

Version history

11 September 2023 1.0.4

Update: Cautions updated.

Droperidol

Mechanism of action

- Droperidol blocks dopamine and alpha receptors centrally, resulting in sedation, reduced agitation and a state of mental detachment, and antiemetic action.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Patients aged greater than or equal to 12 years with acute behavioural disturbance causing a mild to moderate risk to safety, when olanzapine has not been administered or is ineffective.
- ✓ Severe pain, usually in conjunction with an opiate, associated with one of the following:
  - Chronic use of opiates.
  - Severe psychological distress.
  - Nausea and/or vomiting.
- ✓ Management of nausea and agitation or pain that does not respond to an opiate during the last days of life.
- ✓ Nausea and/or vomiting:
  - That persists despite ondansetron, or
  - Where motion sickness is a component of the patient's nausea and/or vomiting, or
  - The patient has cannabinoid hyperemesis syndrome.
- ✓ Vertigo.
- ✓ Primary headache that is severe.

Contraindications

- ✗ Known severe allergy.
- ✗ Aged less than 12 years.
- ✗ Pregnancy (unless indicated for acute behavioural disturbance).

Droperidol

Cautions

- Altered level of consciousness. This will increase and prolong the effects.
- Parkinson's disease. There is a risk of worsening the movement disorder associated with Parkinson's disease.
- Concurrent administration of ketamine or midazolam. This will increase and prolong the effects.
- Aged greater than or equal to 75 years, particularly if frail. The effects of droperidol will be increased and prolonged in this cohort.
- Signs of shock. Droperidol may make shock worse.

Use in pregnancy or when breastfeeding

- The likelihood of a pregnant or breastfeeding patient requiring droperidol for acute behavioural disturbance is very low, however the balance of risk is such that it should be administered if indicated in this context.
- For all other indications, the balance of risk is not in favour of administration of droperidol and therefore it is contraindicated.

Dosage

- For analgesia:
  - 0.625 mg IV/IM for an adult, once only.
- For acute behavioural disturbance:
  - 10 mg IV/IM for an adult.
  - Consider reducing the dose to 5 mg if a caution is present.
  - The dose may be repeated once after 20 minutes.
- For care during the last days of life:
  - 1.25 mg IV or 2.5 mg IM/SC.
  - The dose may be repeated as required.
- For nausea and/or vomiting:
  - 0.625 mg IV/IM for an adult, once only.
- For vertigo:

Droperidol

- 1.25mg IV/IM for an adult, once only.
- Consider reducing the dose to 0.625 mg IV/IM if a caution is present.

- For primary headache that is severe:
  - 1.25 mg IV/IM for an adult, once only.
  - Consider reducing the dose to 0.625 mg IV/IM if a caution is present.

Administration

- IV administration in acute behavioural disturbance:
  - Draw up the required dose and dilute to a total of 10 ml using 0.9% sodium chloride.
  - Administer IV over 1-2 minutes.

- IV administration for nausea and/or vomiting, an adjunct to analgesia, vertigo or primary headache that is severe:
  - Draw up 2.5 mg and dilute to a total of 4 ml using 0.9% sodium chloride. This gives a 0.625 mg/ml solution.
  - Administer the required dose as an IV bolus.

- IM administration: Administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.

- SC administration:
  - Administer undiluted subcutaneously into the abdominal wall.
  - There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.

Common adverse effects

- Hypotension. This may occur if a large IV dose is administered rapidly.

Usual onset of effect

- IV/IM: 5-10 minutes.

Usual duration of effect

- 4-6 hours.

Droperidol

Usual preparation

- Ampoule containing 2.5 mg in 1 ml.

Pharmacokinetics

- Droperidol is predominantly metabolised in the liver with metabolites being excreted in the urine.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Intoxication. Droperidol will have increased sedative effects if the patient is intoxicated with alcohol or has taken recreational drugs.
- Sedative drugs. Concurrent administration with other sedative drugs (such as olanzapine or midazolam) will result in an increased sedative effect.

Additional information

- Droperidol has been reported to prolong the QT interval. This generally involved repeated and/or high doses and one or two doses are safe, even if the patient is known to have a prolonged QT interval.
- Droperidol may cause dyskinesia (abnormal, uncoordinated and involuntary movements) but this is unusual following one or two doses.

More Information

Version history

16 June 2026 1.1.0.1
Update: Indications and dosage updated.

9 June 2025 1.0.5.3
Update: Indications and contraindication updated.

11 September 2023 1.0.4
Update: Indications and dosage updated.

22 November 2022 1.0.1
Update: Aligned with analgesia CPG.

Droperidol

Enoxaparin

Mechanism of action

- Enoxaparin is a low molecular weight heparin (LMWH) anticoagulant.
- Enoxaparin potentiates the activity of antithrombin III (a naturally occurring anticoagulant) causing inhibition of multiple coagulation factors, particularly factor Xa.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ STEMI in conjunction with fibrinolytic therapy.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Clinically significant bleeding. Enoxaparin will increase bleeding.
- ⚠ At risk of bleeding. If there are any cautions present within the fibrinolytic therapy checklist, personnel must seek clinical advice or discuss with the STEMI coordinator prior to administration.
- ⚠ Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of STEMI occurring in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- IV: 30 mg, only if the patient is aged less than 75 years.

Enoxaparin

- SC: Dosage is based on age and known (or estimated) weight.

Weight	Age less than 75 years		Age 75 years or older	
	Enoxaparin (dose SC)	Enoxaparin (volume SC)	Enoxaparin (dose SC)	Enoxaparin (volume SC)
<60 kg	60 mg	0.6 ml	45 mg	0.45 ml
60-69 kg	70 mg	0.7 ml	50 mg	0.5 ml
70-79 kg	80 mg	0.8 ml	60 mg	0.6 ml
80-89 kg	90 mg	0.9 ml	70 mg	0.7 ml
≥90 kg	100 mg	1 ml	75 mg	0.75 ml

Administration

- IV:
  - Dilute 100 mg to a total of 10 ml using 0.9% sodium chloride. This gives a 10 mg/ml solution.
  - Discard 7 ml.
  - Administer 30 mg (3 ml) IV.
- SC:
  - Administer subcutaneously into the abdominal wall.
  - There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.
  - Discard unwanted drug from the syringe before administration.
  - If an error is made in discarding unwanted drug volume and the dose remaining in the syringe is less than planned, administer the remaining dose.

Common adverse effects

- Increased bleeding.

Usual onset of effect

- 10-30 minutes.

Enoxaparin

Usual duration of effect

- 12-24 hours.

Usual preparation

- Pre-filled syringe containing 100 mg in 1 ml.

Pharmacokinetics

- Enoxaparin is predominately excreted in urine.
- Clearance is prolonged if the patient has significant renal impairment, but this does not alter the initial (loading) dose.

Common interactions

- The risk of bleeding will be increased if the patient is taking an anticoagulant.

More Information

Version history

3 August 2026 1.1.0.6
Update: Added IV administration and dosing.

11 September 2023 1.0.4
Update: Cautions updated.

Fentanyl

Mechanism of action

- Fentanyl is an opiate analgesic. It is an opiate agonist (or stimulator) that binds to opiate receptors in the brain and spinal cord causing analgesia.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Moderate to severe pain.
- ✓ Cardiogenic pulmonary oedema with severe anxiety.
- ✓ Rapid sequence intubation.
- ✓ Sedation post intubation.
- ✓ Control of pain, agitation, or shortness of breath during the last days of life.
- ✓ Autonomic dysreflexia, even if the patient cannot feel pain.

Contraindications

- ✗ Known severe allergy.

Fentanyl

Cautions

- Altered level of consciousness. Fentanyl may further reduce the level of consciousness.
- Aged less than one year. Children under the age of one year are at increased risk of respiratory depression following opiate administration.
- Respiratory depression or at high risk of respiratory depression. For example, severe COPD, morbid obesity or on home BiPAP. Such patients may develop respiratory depression following opiate administration.
- Labour. Opiates cross the placenta and may cause drowsiness and/or respiratory depression in the baby, particularly when administered within an hour or two of birth. Discuss administration with the lead maternity carer if possible. Following birth, close observation of the baby is required and personnel must be prepared to treat respiratory depression.
- Concurrent administration of other opiates, ketamine or midazolam. This will increase and prolong the effects.
- Aged greater than or equal to 75 years, particularly if frail. The effects of fentanyl will be increased and prolonged in this cohort.
- Signs of shock. Fentanyl may make shock worse.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy, but fentanyl should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- IV bolus for analgesia:
  - 10-50 mcg every five minutes for an adult, as required. Use a dose at the lower end of the range if the patient is frail or has signs of shock.
  - See the paediatric drug dose tables for a child.
- IV infusion:
  - Start at 1 mcg/kg/hour, up to a maximum of 150 mcg/hour.
  - Administer additional boluses as required, every five minutes.

Fentanyl

- Adjust the infusion rate as required, every 15-20 minutes.

- IV for autonomic dysreflexia:
  - 20-30 mcg increments to a total of 100 mcg.

- IN for analgesia:
  - 100 mcg for an adult.
  - See the paediatric drug dose tables for a child.
  - Halve the dose if the patient is frail or has signs of shock.
  - Repeat as required every 20 minutes.

- IM and SC for analgesia:
  - 50-100 mcg IM/SC for an adult.
  - See the paediatric drug dose tables for a child.
  - Halve the dose if the patient is frail or has signs of shock.
  - Repeat as required, every 20 minutes, up to a maximum of three doses.

- IV for RSI: see the **'RSI'** guideline.

- IV for sedation post intubation: see the **'post intubation'** guideline.

Administration

- The preferred route for administration is IV.

- IV boluses for analgesia:
  - Dilute 100 mcg to a total of 10 ml using 0.9% sodium chloride for an adult or a child whose weight has been rounded up to 30 kg or more. This final solution contains 10 mcg/ml.
  - Dilute 100 mcg to a total of 100 ml using a 100 ml bag of 5% glucose for a child whose weight has been rounded up to 20 kg or less. This final solution contains 1 mcg/ml.

- IN administration:
  - Draw up fentanyl undiluted, placing half of the total dose into two separate 1 ml syringes. When drawing up the first syringe, draw up an additional 0.1 ml of fentanyl over and the above the planned volume and expel this slowly through the mucosal atomiser in order to fill the dead space. This does not need to be done with subsequent doses.
  - Administer fentanyl IN by rapidly injecting one syringe (half of the total dose) into each nostril. Rapid injection is required to achieve a fine mist, which maximises absorption.

Fentanyl

- IM administration: Administer undiluted. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.
- SC administration:
  - Administer undiluted subcutaneously into the abdominal wall.
  - There is no need to sterilise the skin at the site of injection unless the skin is visibly contaminated.
  - Pinch a fold of skin over the anterior abdominal wall between thumb and forefinger. Introduce the entire length of the needle using a dart technique and inject.

Common adverse effects

- Respiratory depression.
- Bradycardia.
- Hypotension.
- Sedation.
- Nausea and vomiting.
- Itch.
- Euphoria.

Usual onset of effect

- IV: 2-5 minutes. The maximal analgesic and respiratory depressant effects may not occur until 10-15 minutes and this may be longer in the elderly.
- IN: 5-10 minutes.
- IM/SC: 5-10 minutes.

Usual duration of effect

- 30-60 minutes.
- The effect on respiration may last for several hours.

Fentanyl

Usual preparation

- Ampoule containing 100 mcg in 2 ml.

Pharmacokinetics

- Fentanyl is more lipophilic (fat soluble) than morphine and this is why fentanyl is well absorbed through the nasal mucosa.
- Fentanyl may cause a small amount of histamine release. In combination with relief of pain this usually results in a small fall in blood pressure.
- Fentanyl is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects will be increased in the presence of other opiates and sedatives, for example, benzodiazepines or alcohol.

Additional information

- A patient administered fentanyl must be given a clear recommendation to be transported to a medical facility by ambulance, unless the patient is being treated using the 'care during the last days of life', 'patella dislocation' or 'shoulder dislocation' guidelines.
- Transport should usually be to an ED unless it is appropriate for the patient to be transported to a primary care facility.
- Fentanyl is usually the preferred opiate if:
  - The patient requires analgesia for a short period of time, or
  - The patient has renal impairment, or
  - The patient has signs of shock.

More Information

Fentanyl

Version history

16 June 2026 1.1.0.1

Update: Dosage updated.

22 November 2022 1.0.1

Update: Contraindications, cautions and aligned with analgesia and end of life CPGs.

Post Intubation

Use this guideline if the patient has been intubated and ETT placement has been confirmed by capnography. If the patient has been intubated during cardiac arrest, this guideline should only be used if sustained ROSC occurs.

- Ventilate using the **'mechanical ventilation'** guideline.
- Administer fentanyl and ketamine:
  - Draw up 200 mcg of fentanyl in combination with 200 mg of ketamine. Dilute with 0.9% sodium chloride to a total of 20 ml. This solution contains 10 mcg/ml of fentanyl and 10 mg/ml of ketamine.
  - Commence an infusion at 0.2 ml/kg/hr using an infusion device, or
  - For an adult administer 5 ml of this solution IV (50 mcg of fentanyl and 50 mg of ketamine) every 15-30 minutes, titrating the frequency to the clinical signs of the patient's level of sedation.
  - For a child administer the volume described in the paediatric drug dose tables IV every 15-30 minutes, titrating the frequency to the clinical signs of the patient's level of sedation.
- Administer rocuronium as required:
  - 100 mg IV for an adult weighing greater than 80 kg.
  - 50 mg IV for an adult weighing 50-80 kg.
  - See the paediatric drug dose tables for a child.
- Administer 0.9% sodium chloride IV if there are signs of hypovolaemia or the patient is hypotensive:
  - 500 ml for an adult.
  - 10 ml/kg for a child.
  - Repeat as required.
- Administer metaraminol IV in addition to 0.9% sodium chloride, if the patient remains hypotensive.
  Metaraminol for an adult:
  - Administer via an infusion device:
    - Administer an initial bolus of 0.5-1 mg metaraminol IV.
    - Start the infusion at 2 mg/hour and adjust the rate as required.
  - Alternatively, administer a bolus of 0.5-1 mg metaraminol IV every 5-10 minutes as required.

  Metaraminol for a child:
  - Administer a bolus of metaraminol IV (see the paediatric drug dose tables) every 5-10 minutes as required.
- Consider administering adrenaline IV if the blood pressure is unresponsive to metaraminol or the patient is hypotensive and bradycardic.
  Adrenaline for an adult:

Post Intubation

- Administer an adrenaline infusion via an infusion device. Start at a rate of 0.5 mg/hour and adjust the rate as required, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 2 drops/second and adjust the rate as required, or
  - Administer a bolus of 0.01 mg adrenaline (10 ml) IV every 1-2 minutes as required.

Adrenaline for a child aged 5-14 years:

- Administer an adrenaline infusion via an infusion device. Start at a rate of 0.25 mg/hour and adjust the rate as required, or
- Place 1 mg of adrenaline into a 1 litre bag of 0.9% sodium chloride (1:1,000,000 solution):
  - Administer as an infusion. Start at 1 drop/second and adjust the rate as required, or
  - Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.

Adrenaline for a child aged less than five years:

- Administer IV boluses (see the paediatric drug dose tables) every 1-2 minutes as required.
- Do not administer adrenaline as an IV infusion.

Additional information

Sedation and neuromuscular blockade

- Provided the patient is not in cardiac arrest, all patients that have been intubated should receive sedation (preferably via an infusion device) and neuromuscular blockade.
- Consider administering sedation without neuromuscular blockade, provided this is safe, if it is considered likely that the patient will receive end of life care within the emergency department.
- When administering sedation:
  - Use doses at the lower end of the described ranges if the patient is small, frail, or cardiovascularly unstable.
  - Titrate the dose and frequency of administration to the clinical signs of the patient's level of sedation. Increase the dose and frequency of administration if the patient has hypertension and tachycardia or shows signs of tears. Decrease the dose and frequency of administration if the patient has hypotension.

Ensuring a safe ETT cuff seal

- It is very easy to generate a high pressure within the ETT cuff and this may cause tracheal injury by impairing the blood supply to the tracheal mucosa. Tracheal injury may lead to tracheal stenosis and this may cause significant morbidity.

Post Intubation

- As soon as feasible, the cuff should be checked to ensure that the cuff pressure is between 20-30 cmH_2O.

More Information

Version history

28 October 2024 1.0.5

Update: Multiple changes.

Glucagon

Mechanism of action

- Glucagon increases the blood glucose level by stimulating glycogenolysis (the breakdown of glycogen into glucose), predominantly within the liver.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Hypoglycaemia when the patient cannot safely swallow glucose/food and IV access cannot be obtained.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- 1 mg IM once for an adult or child aged greater than or equal to five years.
- 0.5 mg IM once for a child aged less than five years.

Administration

- Dissolve the powder using the syringe within the kit and administer IM.
- The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

Glucagon

Common adverse effects

- None.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 15-60 minutes.

Usual preparation

- Ampoule containing 1 mg as powder.

Pharmacokinetics

- Glucagon is predominantly excreted unchanged into bile and urine.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

- None.

Additional information

- Glucagon relies on stored glycogen being available to exert its effect and if glycogen stores are not available, glucagon may be ineffective. Examples include if the patient:
  - Does not have diabetes.
  - Has sepsis.
  - Is a young child.
  - Has undergone strenuous exercise.
  - Has not eaten food for more than 12 hours.
  - Is suffering from adrenal insufficiency.

Glucagon

- Is suffering from chronic hypoglycaemia.
- Is suffering from alcohol-induced hypoglycaemia.

- Following glucagon administration the patient's glycogen stores will be depleted. For this reason it is important the patient eats food as described in the **'hypoglycaemia'** guideline.

- Glucagon also reduces the tone and motility of the smooth muscle in the gastrointestinal tract, but is not recommended in the setting of oesophageal obstruction.

- Glucagon is sometimes suggested as part of the treatment for bradycardia caused by beta-blockers because it stimulates cardiac cells via a mechanism that is independent of the beta receptor. However, glucagon has almost no role in the out-of-hospital setting because it rarely provides a sustained heart rate rise in addition to adrenaline and requires much higher doses than that carried by ambulance personnel.

Glucose Gel

Mechanism of action

- Glucose gel provides a source of glucose that can be easily swallowed and is rapidly absorbed.

Authorised practice levels

- All personnel. ATP is not required to administer glucose gel.
- First Responders: refer to 'diabetic emergencies' in the FR Field Guide.

Indications

- ✓ Hypoglycaemia in adults and children provided the patient is conscious enough to be able to swallow safely.
- ✓ Hypoglycaemia in neonates.

Contraindications

- ✕ None.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- 10-20 g for all ages.
- Administer one sachet and repeat every ten minutes if hypoglycaemia persists or recurs.

Administration

- Administer PO.

Glucose Gel

- Glucose gel may be spread on the gums, tongue and inside of the cheeks of a baby or small child.

Common adverse effects

- None.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 30-60 minutes.

Usual preparation

- There are multiple different brands.
- Most are a sachet containing 10-20 g glucose.

Pharmacokinetics

- Glucose is absorbed in the stomach and small intestine.
- Glucose is rapidly metabolised by cells.

Common interactions

- None.

Additional information

- Document the approximate number of grams of oral glucose administered.

GTN (Glyceryl Trinitrate) IV

Mechanism of action

- GTN is a vasodilator. It acts on vascular smooth muscle to cause venous and arterial vasodilation, with the predominant effect being on veins.
- The mechanism of action is not clear, but it appears that GTN results in the formation of nitric oxide which is a vasodilator. GTN causes:
  - A reduction in venous return (preload) to the heart. This reduces ventricular filling and cardiac output which reduces myocardial oxygen demand.
  - Arterial dilation which reduces peripheral resistance (afterload). This reduces the force the left ventricle must overcome to eject blood into the arteries which reduces myocardial oxygen demand.
  - Dilation of the coronary arteries which may increase coronary blood supply, though this is not usually clinically significant.

Authorised practice levels

- CCPs.

Indications

- ✓ Cardiogenic pulmonary oedema and the patient is not rapidly improving with GTN SL and CPAP.

Contraindications

- ✗ Known severe allergy.
- ✗ Systolic BP less than 110 mmHg.
- ✗ Heart rate less than 40/minute.
- ✗ Heart rate greater than 150/minute.
- ✗ Ventricular tachycardia.

GTN (Glyceryl Trinitrate) IV

Cautions

- STEMI, particularly STEMI involving the right ventricle. GTN may cause a significant fall in cardiac output and if there are signs of low cardiac output GTN should be withheld.
- The patient is frail.
- Signs of shock. The patient's reduced cardiac output may fall further with GTN administration.
- Dysrhythmia. This may cause a reduced cardiac output which may fall further with GTN administration.
- Has taken a phosphodiesterase inhibitor in the last 24 hours. See common interactions.
- Known aortic or mitral stenosis. With aortic or mitral stenosis, cardiac output may be reduced as a result of the narrowed valve and a fall in preload may cause further fall in cardiac output.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring GTN IV is very low, and personnel must seek clinical advice prior to administration.

Dosage

- Start the infusion at 2 mg/hour. Adjust the rate as required by 0.5 mg/hour every 5-10 minutes, titrating to effect.

Administration

- Draw up 50 mg of GTN from a 50 mg/10 ml ampoule and dilute to a total of 50 ml using 0.9% sodium chloride. This solution contains 1 mg/ml.
- Administer via an infusion device.

Common adverse effects

- Hypotension.
- Flushing.
- Headache.

GTN (Glyceryl Trinitrate) IV

- Tachycardia.
- Feeling light-headed.

Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 3-5 minutes following cessation of the infusion.

Usual preparation

- Ampoule containing 50 mg/10 ml.

Pharmacokinetics

- GTN is predominantly metabolised in the liver.

Common interactions

- The effects may be increased if the patient is taking an antihypertensive medicine.
- Severe and/or prolonged hypotension may occur if a phosphodiesterase inhibitor has been taken within the last 24 hours:
  - Phosphodiesterase inhibitors can be used in the treatment of erectile dysfunction and pulmonary hypertension (particularly sildenafil).
  - These medicines are long-acting vasodilators and the administration of GTN may cause further vasodilation.

Additional information

- Particular caution must be used in the presence of STEMI involving the right ventricle:
  - STEMI involving the right ventricle can result in a significant reduction in right ventricular contractility.
  - When the right ventricle is significantly impaired, it may provide little in the way of contribution to cardiac output and blood may be passively flowing down a pressure gradient between the inferior vena cava (IVC), the superior vena cava (SVC) and the left atrium.

GTN (Glyceryl Trinitrate) IV

- This may result in the preload (filling) of the left side of the heart being dependent on the venous pressures within the IVC and SVC.
- GTN can result in a significant fall in venous pressure (and thus a fall in preload) which may cause a significant fall in cardiac output.

More Information

Version history

11 September 2023 1.0.4

New: Guideline released.

Hydrocortisone

Mechanism of action

- Hydrocortisone is a corticosteroid with anti-inflammatory and immunosuppressant actions. It inhibits the production of inflammatory mediators, including prostaglandins and leukotrienes, resulting in a reduction in the inflammatory and immune response.

Authorised practice levels

- Paramedics:
  - At risk of adrenal crisis or suspected to have adrenal crisis.
  - Severe or immediately life-threatening asthma.
  - Severe COPD or imminent respiratory arrest from COPD.
- ICPs and CCPs: All indications.

Indications

- ✓ Adults with angioedema occurring during inter-hospital transfer for stroke clot retrieval.
- ✓ At risk of adrenal crisis or suspected to have adrenal crisis.
- ✓ Severe or immediately life-threatening asthma.
- ✓ Severe COPD or imminent respiratory arrest from COPD.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. However, there is significant clinical experience with steroids, they appear to be safe and hydrocortisone should be administered if indicated.

Hydrocortisone

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 100 mg for an adult.
- See the paediatric drug dose tables for a child.

Administration

- Press down on the plastic cap to force the diluent into the vial.
- Mix the solution by turning the vial upside down several times. Do not shake the vial.
- Remove the central plastic cap and draw up the contents.
- For IV administration:
  - Dilute to approximately 10 ml using 0.9% sodium chloride and administer over 1-2 minutes.
- For IM administration:
  - Do not further dilute after reconstitution. The preferred IM site is the lateral thigh. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Fatigue.
- Sodium and water retention. This may worsen hypertension and heart failure, but is usually only of clinical significance with repeated and/or prolonged dosing.

Usual onset of effect

- 10-20 minutes.

Usual duration of effect

- 8-12 hours.

Hydrocortisone

Usual preparation

- Vial with two chambers, one containing 100 mg of hydrocortisone powder and the other containing 2 ml of sterile water for reconstitution.

Pharmacokinetics

- Hydrocortisone is metabolised in the liver.
- Clearance is prolonged if the patient has significant liver impairment, but this does not alter the initial (loading) dose.

Common interactions

- None.

More Information

Version history

16 June 2026 1.1.0.1
Update: Indications updated.

13 February 2026 1.0.7.1
Correction: Removed advice of shaking vial until dissolved.

9 June 2025 1.0.5.3
Update: Clinical level, indications, dosage and administration information updated.

Ibuprofen

Mechanism of action

- Ibuprofen is a non-steroidal anti-inflammatory drug (NSAID) that is predominantly used for treating pain.
- Ibuprofen inhibits the activity of the enzyme prostaglandin synthetase, reducing prostaglandin production and causing a reduction in inflammation, pain and fever.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to 'pain relief' in the FR Field Guide

Indications

- ✓ Mild to moderate pain, usually in combination with paracetamol.
- ✓ Moderate to severe pain, usually in combination with other medicines.

Contraindications

- ✗ Known severe allergy.
- ✗ Pregnancy.
- ✗ The presence of sepsis, dehydration, shock or clinically significant bleeding. Ibuprofen can worsen renal impairment and increase the risk of bleeding.
- ✗ Known worsening of bronchospasm with NSAIDs. Some patients with asthma or COPD have known worsening of bronchospasm with NSAIDs. If there is a clear history of significant bronchospasm with NSAIDs, ibuprofen must not be administered.

Ibuprofen

Cautions

- The patient has taken ibuprofen within the last four hours. Ibuprofen is contained in many products such as cold and flu tablets/drinks, combination analgesics and migraine tablets. Additional ibuprofen may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold ibuprofen if there is any doubt.
- Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- Age greater than or equal to 75 years, particularly if frail. In this setting, renal impairment is likely and ibuprofen may worsen renal impairment.

Use in pregnancy or when breastfeeding

- May cause foetal harm during pregnancy and should not be administered. In particular, ibuprofen has been associated with premature delivery and premature closure of the ductus arteriosus when administered during the third trimester of pregnancy.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 600 mg for an adult weighing greater than 80 kg.
- 400 mg for an adult weighing 80 kg or less.
- See the paediatric drug dose tables for a child.

Administration

- Administer PO.
- Children unable to swallow tablets may be administered ibuprofen syrup, or tablets that have been crushed and placed in a soft food such as jam or honey.

Common adverse effects

- Renal impairment.

Ibuprofen

- Increased bleeding.
- Although indigestion, gastrointestinal ulceration and gastrointestinal bleeding are commonly listed as adverse effects, these are only associated with chronic administration.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 4-6 hours.

Usual preparation

- 200 mg tablets.
- Syrup containing 20 mg/ml.

Pharmacokinetics

- Ibuprofen is absorbed in the stomach and small intestine. The presence of food in the stomach will delay absorption, but this is not usually clinically significant.
- Ibuprofen is metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Warfarin. Ibuprofen displaces warfarin from binding sites and increases the activity of warfarin.

Additional information

- Ibuprofen is not indicated for pain associated with myocardial ischaemia.
- All personnel (including those without ATP) may give ibuprofen to a patient for self-administration, provided the package instructions are followed.

Ibuprofen

- Ibuprofen is useful in the context of soft tissue pain, musculoskeletal pain and renal colic, particularly if an opiate is not going to be administered.
- Ibuprofen may be administered in addition to other medicines for mild to moderate pain, particularly those not requiring transport or opiate analgesia.
- A patient may be administered ibuprofen for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required.

More Information

Version history

22 November 2022 1.0.1

Update: Indications, contraindications and cautions.

Ipratropium

Mechanism of action

- Ipratropium is a bronchodilator.
- Ipratropium is an anticholinergic agent with predominantly antimuscarinic activity. It antagonises (blocks) acetylcholine receptors, causing vagal inhibition resulting in bronchodilation.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Bronchospasm secondary to asthma or COPD.
- ✓ Prominent bronchospasm secondary to airway burns, smoke inhalation or chest infection.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ipratropium should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 0.5 mg for adults and children once only.

Ipratropium

Administration

- Administer nebulised undiluted (in combination with 5 mg of salbutamol) using:
  - Air as the driving gas for COPD.
  - Oxygen as the driving gas for asthma.

Common adverse effects

- Tachycardia.
- Dry mouth.
- Blurred vision. This usually only occurs with repeated doses.

Usual onset of effect

- 2-5 minutes.

Usual duration of effect

- 6 hours.

Usual preparation

- Ampoule containing 0.5 mg in 2 ml.

Pharmacokinetics

- Only a small amount of the nebulised dose is absorbed, with most of the dose being nebulised to the atmosphere. The inhaled ipratropium is absorbed through the lungs and some is swallowed.
- Excretion is predominantly via the urine.
- Clearance is prolonged if the patient has significant kidney impairment, but this does not alter the initial (loading) dose.

Ipratropium

Common interactions

- None.

Additional information

- Ipratropium does not have a significant role in the treatment of bronchospasm as a result of smoke, toxic gas inhalation or chest infection. However, it may be administered if bronchospasm is prominent.
- Ipratropium has been reported to cause worsening of glaucoma, but only with frequent doses of nebulised ipratropium (with the nebulised drug contacting the eyes) in the setting of poorly controlled glaucoma. Glaucoma is not a caution when administering a single dose.

More Information

Version history

22 November 2022 1.0.1
Update: Aligned with asthma and COPD CPGs.

Ketamine

Mechanism of action

- Ketamine has complex actions but is predominantly an N-methyl-d-aspartate (NMDA) receptor antagonist (blocker), resulting in inhibition of excitatory neurotransmitters in the brain.
- Low doses cause analgesia, larger doses cause amnesia and dissociation, and high doses cause anaesthesia.

Authorised practice levels

- Paramedics: analgesia.
- ICPs and CCPs: all indications.

Indications

- ✓ Severe pain that has not been adequately controlled with an opiate.
- ✓ Inducing dissociation.
- ✓ Acute behavioural disturbance causing a severe to immediately life-threatening risk to safety.
- ✓ Rapid sequence intubation (RSI).
- ✓ Significant movement during CPR that is interfering with resuscitation.
- ✓ Asthma with severe agitation that is impairing the ability to safely provide treatment or transport.

Contraindications

- ✗ Known severe allergy.

Ketamine

Cautions

- Altered level of consciousness. Ketamine may reduce the level of consciousness.
- Signs of shock. Ketamine may make shock worse.
- Concurrent administration of opiates or midazolam. This will increase and prolong the effects.
- Aged greater than or equal to 75 years, particularly if frail. The effects of ketamine will be increased and prolonged in this cohort.
- Aged less than five years.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ketamine should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- For analgesia:
  - IV:
    - 0.2 mg/kg IV up to a maximum of 20 mg for an adult, administered over approximately 15 minutes. Repeat as required.
    - Consider commencing an infusion using service guidelines.
    - See the paediatric drug dose tables for a child and administer IV over approximately 15 minutes. Repeat as required.
  - IM or PO:
    - 0.5 mg/kg IM or PO for an adult, up to a maximum of 50 mg, if IV access cannot be obtained. This may be repeated once after 20 minutes.
    - See the paediatric drug dose tables for a child. This may be repeated once after 20 minutes.
- For dissociation:
  - 1 mg/kg IV (up to a maximum of 150 mg) for an adult. This may be repeated once after five minutes.
  - 5 mg/kg IM (up to a maximum of 400 mg) for an adult, once only.
  - See the paediatric drug dose tables for a child.
  - Halve all doses if a caution is present.

Ketamine

- For acute behavioural disturbance:
  - 1 mg/kg IV (up to a maximum of 150 mg) every five minutes as required, or
  - 5 mg/kg IM, rounded to the nearest 100 mg (up to a maximum of 400 mg).
  - The IM dose may be repeated once after 20 minutes.
- For asthma with severe agitation: 0.5 mg/kg IV, up to a maximum of 50 mg. This may be repeated every 10 minutes as required.
- For RSI: see the **'RSI'** guideline.
- For significant movement during CPR that is interfering with resuscitation administer 1 mg/kg (up to a maximum of 150 mg) of ketamine. This may be repeated once if required.

Administration

- The preferred route of administration is IV.
- Administer analgesia doses IV over approximately 15 minutes as this appears to reduce adverse side effects.
- For analgesia: Dilute 100 mg (1 ml) of ketamine in a 100 ml bag of 5% glucose. Shake well and label:
  - For adults: draw up the dose and administer 2 mg (2 ml) every 1-2 minutes.
  - For children: draw up the dose, dilute the volume further to a total of 20 ml using 0.9% sodium chloride and administer 2 ml every 1-2 minutes.
- Administer dissociation doses by diluting 100 mg (1 ml) to a total of 10 ml using 0.9% sodium chloride and administering the dose as an IV bolus.
- The IM route is preferred over the PO route as IM absorption is more reliable. Administer IM ketamine undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.
- The PO route should be reserved for the very unusual circumstance in which IM injection is contraindicated or not feasible. Administer ketamine PO undiluted in a liquid, for example paracetamol syrup or water.

Common adverse effects

- Transient hypertension.
- Tachycardia.

Ketamine

- Apnoea.
- Nausea and vomiting.
- Sedation.
- Hallucinations.

Usual onset of effect

- IV: 1-2 minutes.
- IM: 5-10 minutes.
- PO: 10-20 minutes.

Usual duration of effect

- 10-60 minutes.

Usual preparation

- Ampoule containing 200 mg in 2 ml.

Pharmacokinetics

- Ketamine is predominantly metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects will be increased in the presence of other analgesic medicines or sedatives, for example opiates, benzodiazepines or alcohol.

Additional information

- Warn the patient they may feel strange following ketamine administration.

Ketamine

- Prior to administering ketamine for pain, sufficient opiates should be administered so that further doses are not providing additional analgesia. This will usually require 150-200 mcg of fentanyl or 15-20 mg of morphine for an adult.
- Do not treat hallucinations routinely with midazolam because the combination of midazolam and ketamine is commonly associated with a reduced level of consciousness, particularly if an opiate has also been administered. Most hallucinations will settle with a combination of explanation and time. However, midazolam in low doses may be administered IV if the hallucinations are severe provided the patient is obeying commands and physiologically stable.
- Round the patient's weight up to the nearest 10 kg when calculating doses.

More Information

Version history

16 June 2026 1.1.0.1
Update: Contraindications, cautions and dosage updated.

11 September 2023 1.0.4
Update: Multiple changes.

22 November 2022 1.0.1
Update: Indications, cautions, infusion instructions and aligned with analgesia CPG.

Labetalol

Mechanism of action

- Labetalol is a beta-blocker and an alpha-blocker. It antagonises (blocks) beta-1 receptors in the heart, causing a decrease in heart rate, cardiac output and blood pressure and it antagonises (blocks) alpha-1 receptors peripherally, causing vasodilation and a decrease in blood pressure.

Authorised practice levels

- ICPs and CCPs.
- Paramedics may only administer labetalol after seeking clinical advice.

Indications

- ✓ Control of hypertension prior to fibrinolytic therapy for STEMI.
- ✓ Control of hypertension during inter-hospital transfer for STEMI.
- ✓ Control of hypertension during inter-hospital transfer for stroke clot retrieval.
- ✓ Control of hypertension associated with autonomic dysreflexia.
- ✓ Control of hypertension associated with eclampsia.
- ✓ Control of hypertension during inter-hospital transfer for acute aortic dissection.

Contraindications

- ✗ Known severe allergy.
- ✗ Bradycardia. Labetalol will further reduce the heart rate.
- ✗ Hypotension. Labetalol will further reduce the blood pressure.

Labetalol

Cautions

- 1st degree heart block. Labetalol may cause bradycardia.
- Known sick sinus syndrome without an internal pacemaker in place. Labetalol may cause bradycardia.
- Previous 2nd or 3 rd degree heart block without an internal pacemaker in place. Labetalol may cause worsening of heart block and bradycardia.
- Asthma or COPD. Labetalol may cause bronchospasm and should usually be withheld if the patient regularly takes bronchodilators.
- Heart failure. Labetalol will reduce cardiac output and may make heart failure worse.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

Dosage

- Administer a 20 mg bolus IV every five minutes, as required, while setting up an infusion.
- Commence an IV infusion at 1 mg/kg/hour, up to a maximum of 150 mg/hour.
- Adjust the rate as required by 20 mg/hour every 10 minutes, titrating to effect.
- Administer additional boluses of 20 mg IV every five minutes as required.

The table below provides an overview of the blood pressure threshold for commencing treatment and the BP maintenance goal.

Indication	BP to commence treatment	BP maintenance goal
Prior to fibrinolytic therapy for STEMI	SBP > 180 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.
IHT for STEMI	SBP > 180 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.

Labetalol

IHT for SCR	SBP > 180 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.
Autonomic dysreflexia	SBP > 160 mmHg, or SBP > 40 mmHg above the patient's normal.	SBP 140-160 mmHg, and DBP < 110 mmHg.
Eclampsia	SBP > 160 mmHg, or DBP > 110 mmHg.	SBP 140-160 mmHg, and DBP < 110 mmHg.
IHT for acute aortic dissection	SBP > 120 mmHg.	SBP 100-120 mmHg.

Administration

- Draw up 100 mg undiluted and administer 20 mg as a bolus, before commencing an infusion via an infusion device.

Common adverse effects

- Hypotension.
- Bradycardia.
- Bronchospasm.

Usual onset of effect

- 2-3 minutes.

Usual duration of effect

- 1-2 hours.

Usual preparation

- Ampoule containing 100 mg in 20 ml.

Labetalol

Pharmacokinetics

- Labetalol is metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The blood pressure effect will be potentiated by other medicines that lower blood pressure. For example, GTN, antihypertensive medicines and amiodarone.
- The heart rate effects will be potentiated by other medicines that lower heart rate. For example, amiodarone and centrally acting calcium channel blockers such as diltiazem.

More Information

Version history

16 June 2026 1.1.0.1
Update: Indications and dosage updated.

9 June 2025 1.0.5.3
Update: Indications and dosage updated.

11 September 2023 1.0.4
Update: Dosage and additional information updated.

Levetiracetam

Mechanism of action

- Levetiracetam is an anticonvulsant. It is thought to inhibit seizure activity by blocking some calcium channels and also binding synaptic proteins to modulate neurotransmitter release.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Seizure that continues or recurs after two doses of parenteral midazolam.
- ✓ Seizure activity associated with severe TBI, even if seizure activity has ceased following midazolam.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- 2 g IV for an adult, this may be repeated once if the seizure continues or recurs ten minutes after the first dose.
- See the paediatric drug dose tables for a child. This may be only administered once.

Administration

- Draw up the required dose.

Levetiracetam

- Administer this IV over 5 minutes preferably into a running line.
- Do not administer IM.

Common adverse effects

- None.

Usual onset of effect

- 10 minutes.

Usual duration of effect

- 6-8 hours.

Usual preparation

- Ampoule containing 500 mg in 5 ml.

Pharmacokinetics

- Levetiracetam is excreted by the kidneys.
- There are no significant effects from renal impairment on the initial (loading) dose.

Common interactions

- None.

More Information

Version history

9 June 2025 1.0.5.3
Update: Indications updated.

28 October 2024 1.0.5
Update: Indications updated.

Levetiracetam

11 September 2023 1.0.4

New: Guideline released.

Loratadine

Mechanism of action

- Loratadine is a non-sedating antihistamine.
- Loratadine antagonises (blocks) peripheral histamine receptors, blocking the action of histamine and reducing itching and redness.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Minor allergic reactions confined to skin involvement.
- ✓ Itch or rash associated with anaphylaxis, provided all systemic signs of anaphylaxis are improving.

Contraindications

- ✕ Known severe allergy.
- ✕ Age less than one year.

Cautions

- ⚠ Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. Because minor allergic reactions rarely require specific treatment, the balance of risk is such that loratadine should usually be withheld.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 20 mg for an adult or child aged greater than or equal to 12 years.

Loratadine

- 10 mg for a child aged 1-11 years.

Administration

- Administer PO.
- Always ask a parent (or guardian) if a young child can swallow tablets. Loratadine may be crushed and placed in a soft food such as jam or honey.

Common adverse effects

- None.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 12-24 hours.

Usual preparation

- 10 mg tablets.

Pharmacokinetics

- Loratadine is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.
- Increases in plasma concentrations of loratadine have been reported after concurrent administration of ketoconazole, erythromycin and roxithromycin. In a small number of patients this was associated with a prolonged QT interval, but this only occurred with long-term administration.

Loratadine

Additional information

- All personnel (including those without ATP) may give loratadine to a patient for self-administration, provided the package instructions are followed.

More Information

Version history

28 October 2024 1.0.5

Update: Dosage updated.

Magnesium Sulphate

Mechanism of action

- Magnesium is the active ingredient in magnesium sulphate.
- Magnesium reduces bronchial smooth muscle contraction resulting in bronchodilation.
- Magnesium is thought to have multiple mechanisms when treating eclampsia, including vasodilation, blood-brain barrier protection, reduction of cerebral oedema, and central anticonvulsant effects.
- Magnesium has been shown to stablise the cardiac membrane during torsades de pointes.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Bronchospasm secondary to severe or immediately life-threatening asthma.
- ✓ Bronchospasm secondary to severe COPD or imminent respiratory arrest from COPD.
- ✓ If torsades de pointes is present or has occurred.
- ✓ Suspected eclampsia.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Hypotension. Magnesium is a vasodilator and may make hypotension worse.

Use in pregnancy or when breastfeeding

- Safe and should be administered when indicated.

Magnesium Sulphate

Dosage

- Asthma:
  - 10 mmol (2.5 g) IV for an adult.
  - See the paediatric drug dose tables for a child.
  - A second dose may be administered if transport time is longer than 30 minutes and the patient is not improving.
- COPD:
  - 10 mmol (2.5 g) IV for an adult.
  - A second dose may be administered if transport time is longer than 30 minutes and the patient is not improving.
- Torsades de pointes is present or has occurred:
  - 10 mmol (2.5 g) IV for an adult.
  - Repeat once if episodes of torsades de pointes continue.
- Suspected eclampsia:
  - 20 mmol (5 g) IV.

Administration

- Cardiac arrest if torsades de pointes is present or has occurred:
  - Dilute to a total of 10 ml using 0.9% sodium chloride and administer over 1-2 minutes.
- All other indications:
  - Administer IV over approximately 15 minutes.
  - Draw up the dose and dilute to a total of 10 ml using 0.9% sodium chloride (if required) and administer 1 ml every 1-2 minutes, preferably into a running line.
  - Alternatively, in adults or children weighing over 40 kg only, add to a 100 ml bag of 5% glucose, shake well, label and infuse over approximately 15 minutes:
    - 2-3 drops per second via a standard IV administration set will deliver 100 ml over approximately 15 minutes.
    - The administration set will need to be flushed with 0.9% sodium chloride to ensure all the magnesium has been administered.

Magnesium Sulphate

Common adverse effects

- Flushing. Particularly if administered rapidly.
- Hypotension. Particularly if administered rapidly.
- Muscle weakness. This is usually only seen with doses exceeding 20 mmol.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 30-60 minutes.

Pharmacokinetics

- Magnesium is primarily excreted in the urine.
- There are no significant effects from renal impairment on the initial (loading) dose.

Usual preparation

- 5 ml ampoule containing 10 mmol (2.5 g) of magnesium.

Common interactions

- None.

Additional information

- Do not administer magnesium into an IV line that has an adrenaline infusion running through it concurrently, because precipitation may occur.
- Rarely, magnesium may have a role in the treatment of severe pre-eclampsia. In this setting seek clinical advice.
- Document the dose administered in mmol.

Magnesium Sulphate

More Information

Version history

9 June 2025 1.0.5.3

Update: Dosage updated.

11 September 2023 1.0.4

Update: Indications have been added. Dosage, administration and additional information has been updated.

22 November 2022 1.0.1

Update: Indication for COPD.

Metaraminol

Mechanism of action

- Metaraminol is an alpha receptor agonist. It stimulates alpha-1 receptors in peripheral blood vessels, causing vasoconstriction and a rise in blood pressure.
- Some references describe a small degree of stimulation of beta receptors, but this is usually clinically insignificant.

Authorised practice levels

- ICPs and CCPs.

Indications

- ✓ Hypotension in the setting of septic shock, post cardiac arrest, cardiogenic shock, adrenal crisis, severe traumatic brain injury, neurogenic shock, rapid sequence intubation and post intubation.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Bradycardia. Metaraminol may make bradycardia worse as reduced endogenous release of adrenaline may occur as blood pressure is raised. Consider administering adrenaline if the patient is hypotensive and bradycardic.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated. The likelihood of a pregnant or breastfeeding patient requiring metaraminol is very low, but it should be administered if indicated.

Dosage

- Titrate the dose and frequency to the patient's blood pressure:
  - » Administer an infusion via an infusion pump, or
  - » Administer 0.5-1 mg IV every 5-10 minutes as required for an adult, or

Metaraminol

- See the paediatric drug dose tables for bolus doses for a child, and administer IV every 5-10 minutes as required.

- If an infusion is administered in an adult:
  - Administer an initial bolus of 0.5-1 mg IV.
  - Start the infusion at 2 mg/hour and adjust the rate as required.
  - The patient is likely to require approximately 2-3 mg/hour.

Administration

- IV bolus administration in an adult:
  - Dilute 10 mg of metaraminol to a total of 10 ml using 0.9% sodium chloride. This final solution contains 1 mg/ml.
  - Administer the required dose, preferably via a running line.

- IV bolus administration in a child:
  - Draw up 1 ml of metaraminol from a 10 mg/1 ml ampoule.
  - Place into a 100 ml bag of 5% glucose.
  - Shake well and label. This final solution contains 0.1 mg/ml.
  - See the paediatric drug dose tables for dosing/volume and draw up doses from the bag of 5% glucose.

- IV infusion:
  - Follow service guidelines for dilution and infusion pump set up.
  - Adjust the infusion rate as required.

Common adverse effects

- Hypertension. This usually only occurs when boluses greater than 1 mg are administered in an adult.

- Bradycardia. See cautions.

Usual onset of effect

- 1-2 minutes.

Metaraminol

Usual duration of effect

- 10-15 minutes.

Usual preparation

- Ampoule containing 10 mg in 1 ml.
- Pre-filled syringe containing 10 mg in 10 ml.

Pharmacokinetics

- Metaraminol is predominantly metabolised in the liver with metabolites being excreted in the urine.
- Liver impairment does not alter the dose which is titrated to effect.

Common interactions

- None.

Additional information

- Some references describe the IM administration of metaraminol but this has no role in the out-of-hospital setting.

More Information

Version history

9 June 2025 1.0.5.3
Update: Indications updated.

22 November 2022 1.0.1
Update: Usual preparation and administration.

Methoxyflurane

Mechanism of action

- Methoxyflurane is an inhalational analgesic.
- The mechanism of action is not clear.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Acute moderate to severe pain if:
  - An opiate cannot be administered because of a contraindication or caution, or
  - Opiate administration will be significantly delayed, or
  - There will be severe worsening of pain during a procedure or extrication.

Contraindications

- ✕ Known severe allergy.
- ✕ Personal or family history of malignant hyperthermia.
- ✕ Known renal impairment.
- ✕ 15 ml methoxyflurane (5 doses) administered within the last week. Frequent administration increases the risk of renal impairment.

Cautions

- ⚠ Age greater than or equal to 75 years, particularly if frail.
- ⚠ Pre-eclampsia. In this setting, renal impairment is likely and methoxyflurane may worsen renal impairment.
- ⚠ Administration within a confined space.
- ⚠ Acute exacerbation of chronic pain.

Methoxyflurane

Use in pregnancy or when breastfeeding

- Safety has not been formally demonstrated in pregnancy, but methoxyflurane may be administered. Methoxyflurane has been extensively used during labour in Australia and New Zealand for many years without adverse effects.
- Methoxyflurane may cause temporary drowsiness in the baby and administration should be discussed with the lead maternity carer if there are known signs of foetal distress.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- Maximum of 6 ml (two doses) for a patient aged greater than or equal to 12 years.
- Maximum of 3 ml (one dose) for a child aged less than 12 years.

Administration

- Whenever possible, have the patient self-administer methoxyflurane.
- Administer 3 ml (one dose) at a time and always use the charcoal filter. Using a charcoal filter maximises the amount of exhaled methoxyflurane that is absorbed, limiting exposure to personnel.
- Instruct the patient to breathe out through the inhaler.
- Do not administer supplementary oxygen via the inhaler as this significantly increases the amount of methoxyflurane lost through evaporation.
- Place the inhaler in a closed plastic bag if the methoxyflurane has not been fully used. It may subsequently be reused by the same patient.
- An ambulance is not considered a confined space. Maximising ventilation (for example, having ventilation fans on) reduces occupational exposure. Consider not administering methoxyflurane in an ambulance if the patient cannot cooperate with breathing out through the inhaler.

Common adverse effects

- Sedation.
- Feeling light-headed.

Methoxyflurane

Usual onset of effect

- 1-2 minutes.

Usual duration of effect

- 2-5 minutes after stopping administration.

Usual preparation

- 3 ml bottle accompanying a plastic inhaler.

Pharmacokinetics

- Approximately 20% is exhaled. The remainder is metabolised in the liver.
- One of the metabolites is fluoride ions. High concentrations of fluoride ions have been associated with renal impairment and this is the reason for known renal impairment being a contraindication and for having a maximum dose.

Common interactions

- The effects will be increased in the presence of other pain relieving medicines or sedatives, for example opiates, benzodiazepines or alcohol.

Additional information

- Malignant hyperthermia (MH) is a rare, inherited disorder of muscle metabolism affecting approximately 20 families in New Zealand, many of whom are in the Manawatu area. When exposed to some anaesthetic agents (including methoxyflurane), the patient may develop a life-threatening hypermetabolic state with severe hyperthermia. Patients with MH (or a family history of MH) usually know about it.
- Renal failure with dialysis is not a contraindication or a caution to methoxyflurane administration because once a patient is receiving dialysis further renal impairment is of no clinical consequence.
- Kidney stones and/or renal colic are not a contraindication or a caution to methoxyflurane administration because these are rarely associated with renal impairment.
- Methoxyflurane must not be left with a patient who is not being transported to a medical facility by ambulance.

Methoxyflurane

More Information

Version history

16 June 2026 1.1.0.1

Update: Indications updated.

11 September 2023 1.0.4

Update: A new caution has been added (acute exacerbation of chronic pain).

22 November 2022 1.0.1

Update: Contraindications, cautions and aligned with analgesia CPG.

Metoprolol Tartrate

Mechanism of action

- Metoprolol is an immediate release beta blocker. It antagonises (blocks) beta-1 receptors in the heart, causing a decrease in heart rate, cardiac output and blood pressure.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- Adults with moderate cardiovascular compromise as a result of atrial fibrillation, where the ventricular rate is predominantly greater than 130/minute.

Contraindications

- Known severe allergy.
- Hypotension. Metoprolol will further reduce the blood pressure.

Cautions

- First degree heart block. Metoprolol may cause bradycardia.
- Known sick sinus syndrome without an internal pacemaker in place. Metoprolol may cause bradycardia.
- Previous second or third degree heart block without an internal pacemaker in place. Metoprolol may cause worsening of heart block and bradycardia.
- Asthma or COPD. Metoprolol may cause bronchospasm and should usually be withheld if the patient regularly takes bronchodilators.
- Heart failure. Metoprolol will reduce cardiac output and may make heart failure worse.
- Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. The likelihood of administration being required in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.

Metoprolol Tartrate

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 25 mg.

Administration

- Administer PO.

Common adverse effects

- None.

Usual onset of effect

- 10-20 minutes

Usual duration of effect

- 4-6 hours

Usual preparation

- 50 mg tablets.

Pharmacokinetics

- Metoprolol is absorbed within the gastrointestinal tract and metabolised in the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The blood pressure effect will be potentiated by other medicines that lower blood pressure. For example, GTN, antihypertensive medicines and amiodarone.

Metoprolol Tartrate

- The heart rate effects will be potentiated by other medicines that lower the heart rate. For example, amiodarone and centrally acting calcium channel blockers such as diltiazem.

Additional information

- Patients with chronic or paroxysmal atrial fibrillation may be prescribed a beta blocker. Personnel should still administer metoprolol as indicated, even if the patient has taken their own beta blocker.

More Information

Version history

16 June 2026 1.1.0.1

Update: Indications and dosage updated.

11 September 2023 1.0.4

New: Guideline released.

Midazolam

Mechanism of action

- Midazolam is a benzodiazepine.
- Midazolam enhances the activity of gamma-aminobutyric acid (GABA) at GABA receptors within the central nervous system, resulting in anticonvulsant activity, sedation, amnesia, anxiolysis and muscle relaxation.

Authorised practice levels

- Paramedics:
  - IM for acute behavioural disturbance, seizures and care during the last days of life.
  - IV for seizures and care during the last days of life.
- ICPs and CCPs: all routes and all indications.

Indications

- ✓ Generalised seizures that continue for more than five minutes, or seizures are recurrent.
- ✓ Acute behavioural disturbance causing a mild to moderate risk to safety and droperidol is unavailable or ineffective.
- ✓ Pain associated with severe muscle spasm or severe anxiety if adequate analgesia is not being achieved with an opiate and ketamine is not appropriate.
- ✓ Sedation, for example, for joint relocation.
- ✓ Severe anxiety associated with COPD.
- ✓ Control of anxiety or shortness of breath that does not respond to an opiate during the last days of life.

Contraindications

- ✗ Known severe allergy.
- ✗ Aged less than 12 years (analgesia only).

Midazolam

Cautions

- ⚠ Altered level of consciousness. Midazolam will further reduce the level of consciousness.
- ⚠ Respiratory depression or at high risk of respiratory depression. For example, severe COPD, morbid obesity or on home BiPAP. Such patients may develop respiratory depression following midazolam administration.
- ⚠ Signs of shock. Midazolam may worsen shock.
- ⚠ Concurrent administration of opiates or ketamine. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of midazolam will be increased and prolonged in this cohort.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but midazolam should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- The dose of midazolam is dependent on the indication and the route. See the individual guidelines.
- If a dose range is described, use a dose at the lower end of the range if a caution is present.
- If midazolam is being administered as an adjunct to analgesia or sedation, the patient must be rousable at all times.

Administration

- IV administration:
  - Dilute 2 ml from a 15 mg/3 ml ampoule to a total of 10 ml using 0.9% sodium chloride. This solution contains 1 mg/ml.
  - Administer IV as a bolus.
- IM administration:
  - Draw up the dose from a 15 mg/3 ml ampoule. Do not dilute.

Midazolam

- The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Sedation.
- Respiratory depression.
- Hypotension.
- Amnesia.

Usual onset of effect

- IV: 2-3 minutes.
- IM: 3-5 minutes.

Usual duration of effect

- 30-60 minutes. The sedative effect may be longer, particularly in the elderly.

Usual preparation

- Ampoule containing 15 mg in 3 ml.

Pharmacokinetics

- Midazolam is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The effects will be increased and prolonged in the presence of other sedatives or pain relieving medicines (for example, other benzodiazepines, opiates, ketamine and alcohol).

Midazolam

More Information

Version history

9 June 2025 1.0.5.3

Update: Indications updated.

11 September 2023 1.0.4

Update: Indications have been updated to align with changes to EAS CPG 5.1 and 5.8.

22 November 2022 1.0.1

Update: Contraindications, cautions and aligned with analgesia and end of life CPGs.

Morphine

Mechanism of action

- Morphine is an opiate analgesic. It is an opiate agonist that binds to opiate receptors in the brain a spinal cord causing analgesia.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Moderate to severe pain.
- ✓ Cardiogenic pulmonary oedema with severe anxiety.
- ✓ Control of pain, agitation, or shortness of breath during the last days of life.
- ✓ Autonomic dysreflexia, even if patient cannot feel pain.

Contraindications

- ✗ Known severe allergy.
- ✗ Renal failure.

Naloxone

Mechanism of action

- Naloxone is an opiate receptor antagonist (blocker). By blocking opiate receptors, naloxone reverses the effects of opiates, particularly respiratory depression and sedation.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Opiate poisoning is suspected and the patient has a significantly impaired level of consciousness or significantly impaired breathing.
- ✓ Excess adverse effects from administration of opiates.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Chronic opiate use. If the patient is taking an opiate chronically, there is a risk of adverse physiological effects associated with rapid opiate withdrawal.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy, but naloxone should be administered if indicated.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 0.1-0.4 mg IV every five minutes as required for an adult.
- 0.8 mg IM as required for an adult. This may be repeated every 10 minutes.
- See the paediatric drug dose tables for a child.

Naloxone

Administration

- The preferred route for administration is IV.
- IV: dilute 0.4 mg to a total of 4 ml. This final solution contains 0.1 mg/ml.
- Administer the minimum dose required to produce improvement. Rapid reversal of opiates may be associated with seizures, hypertension, pulmonary oedema or severe agitation, particularly if the patient takes opiates regularly.
- IM: administer undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Sweating.
- Tachycardia.
- Hypertension.

Usual onset of effect

- IV: 1-2 minutes.
- IM: 5-10 minutes.

Usual duration of effect

- 30-60 minutes.
- The duration of action of naloxone may be shorter than the duration of action of the opiate that has been administered/taken and naloxone may need to be repeated.

Usual preparation

- Ampoule containing 0.4 mg in 1 ml.

Naloxone

Pharmacokinetics

- Naloxone is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.

Additional information

- There is no role for naloxone in the treatment of cardiac arrest associated with opiate poisoning. In this setting cardiac arrest is secondary to respiratory arrest and once cardiac arrest has occurred naloxone has no useful effect. The best treatment is CPR that includes a focus on ventilation. If ROSC occurs, naloxone should still not be administered because it may be associated with seizures, hypertension, pulmonary oedema or severe agitation.
- There is no evidence to support the commonly held view that adequate oxygenation prior to naloxone administration reduces the severity of agitation following naloxone administration. However, treatment of severe hypoxia takes precedence over the administration of naloxone.
- Acute opiate withdrawal following naloxone administration can lead to patient agitation and/or aggression, particularly when there has also been a period of hypoxia. Ensure a planned team approach to maintain safety of both the patient and personnel.

Olanzapine

Mechanism of action

- Olanzapine is an atypical antipsychotic.
- Olanzapine has actions at multiple receptors within the brain causing a reduction in agitation, sedation, anxiolysis and stabilisation of mood.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Patients aged greater than or equal to 12 years with acute behavioural disturbance causing a mild to moderate risk to safety, when the patient will take an oral medicine.

Contraindications

- ✕ Known severe allergy.
- ✕ Poisoning with an antipsychotic, for example quetiapine, risperidone or olanzapine.

Cautions

- ⚠ Pregnancy.
- ⚠ Intoxication. This will increase and prolong the effects.
- ⚠ Aged greater than or equal to 75 years, particularly if frail. The effects of olanzapine will be increased and prolonged in this cohort.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy and the balance of risk is such that olanzapine should usually be withheld.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Olanzapine

Dosage

- 10 mg PO.
- Consider reducing the dose to 5 mg PO if a caution is present.
- The dose may be repeated once after 20 minutes.

Administration

- Administer PO. The tablet is dispersible and will dissolve in the mouth. A sip of liquid will aid absorption but is not routinely required.
- The tablet may be dissolved in liquid. For example, water, tea or coffee.

Common adverse effects

- Sedation.

Usual onset of effect

- 10-20 minutes.

Usual duration of effect

- 12-24 hours.

Usual preparation

- 5 mg dispersible tablets (wafers).

Pharmacokinetics

- Olanzapine is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Olanzapine

Common interactions

- Intoxication. Olanzapine will have increased and prolonged effects if the patient is intoxicated with alcohol or has taken recreational drugs.
- Sedative drugs. Concurrent administration with other sedative drugs (such as midazolam or droperidol) will result in an increased and prolonged effect.

More Information

Version history

11 September 2023 1.0.4
Update: Indications updated.

22 November 2022 1.0.1
Update: Indications.

Ondansetron

Mechanism of action

- Ondansetron is an antiemetic.
- Ondansetron antagonises (blocks) serotonin receptors centrally in the brain and peripherally in the gastrointestinal tract, resulting in a reduction in nausea and vomiting.

Authorised practice levels

- EMTs: IM.
- Paramedics, ICPs and CCPs: IV and IM.

Indications

- ✓ Clinically significant nausea and/or vomiting.

Contraindications

- ✕ Known severe allergy.
- ✕ Age less than one year.
- ✕ Suspected serotonin syndrome.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy, but ondansetron may be administered if nausea and/or vomiting is severe.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Ondansetron

Dosage

- 8 mg IV once for an adult. See the paediatric drug dose tables for a child.
- 4 mg IM for an adult, if IV access cannot be obtained. See the paediatric drug dose tables for a child.
- The IM dose may be repeated once after 20 minutes if required.
- One IV dose may be administered in addition to one IM dose, if clinically significant nausea and/or vomiting persists and IV access is subsequently obtained. For example, in this setting a maximum total dose of 12 mg may be administered to an adult.

Administration

- The preferred route of administration is IV.
- Administer IV undiluted.
- Administer IM undiluted. The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.

Common adverse effects

- Headache.
- Flushing.

Usual onset of effect

- IV: 2-5 minutes.
- IM: 5-10 minutes.

Usual duration of effect

- 4-8 hours.

Usual preparation

- Ampoule containing 4 mg in 2 ml.

Ondansetron

- Ampoule containing 8 mg in 4 ml.

Pharmacokinetics

- Ondansetron is predominantly metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- Ondansetron has been reported to prolong the QT interval, particularly if high doses are administered in conjunction with other medicines that also prolong the QT interval, for example erythromycin. However, one or two doses in this setting is safe.

Additional information

- Prophylactic administration of ondansetron is not routinely required. Consider administering ondansetron if the nature of the patient's injuries and transport position are such that vomiting would be particularly problematic.
- Ondansetron should not be administered for vomiting associated with an altered level of consciousness because it is rarely effective in this setting.
- Ondansetron has been reported to further prolong the QT interval in patients known to have a prolonged QT syndrome. This generally involved repeated and/or high doses and one or two doses is safe, even if the patient is known to have a prolonged QT syndrome.
- Nausea and/or vomiting in the context of serotonin syndrome must not be treated with ondansetron as this may worsen serotonin syndrome.

More Information

Version history

11 September 2023 1.0.4

Update: Additional information updated to include serotonin syndrome.

Oxygen Administration

- Few guidelines contain specific instructions on oxygen administration and clinical judgement is required.
- Oxygen should usually only be administered if the patient has an SpO_2 less than 92% on air, airway obstruction, or a specific indication as described within these CPGs.
- Use the simplest device and lowest flow rate required to achieve an SpO_2 of 92-96% (or 88-92% if at risk of hypercarbia) unless specific devices and/or flow rates are described in individual guidelines.
- If pulse oximetry is unreliable or unavailable, administer oxygen as appropriate for the patient's clinical condition.
- The oxygen flow rates usually used are:
  - Nasal prongs 1-4 litres/minute.
  - Simple mask 6 litres/minute.
  - Nebuliser mask 8 litres/minute.
  - Reservoir mask 10 litres/minute.
  - Manual ventilation bag 10 litres/minute.
- If a manual ventilation bag is being used to provide ventilation, use the following PEEP settings:
  - Adult or child:
    - Do not attach PEEP during cardiac arrest.
    - Apply PEEP at 5 cmH_2O for all other conditions. CCPs may choose to increase PEEP if clinically indicated.
    - Refer to the **'cardiogenic pulmonary oedema'** guideline if the patient is an adult and has cardiogenic pulmonary oedema.
  - Neonate:
    - Apply PEEP at 5 cmH_2O, including during cardiac arrest.

Additional information

Oxygen administration

- Oxygen is a treatment for hypoxia and not a general treatment for patients that are ill or injured.
- Oxygen administration is restricted to those patients that have an indication to receive it, because when oxygen levels within blood are higher than normal:
  - Blood vessels (particularly small arteries) vasoconstrict. This has the potential to lower blood flow to tissues and organs.
  - Inflammation is increased and this may worsen inflammatory states.

Oxygen Administration

- Routine oxygen administration during neonatal resuscitation appears to make outcomes worse. This is why oxygen is reserved for deterioration despite initial ventilation.

Delivery devices

- For most patients, nasal prongs or a simple mask is sufficient.
- Reservoir masks should be reserved for patients requiring a high concentration of inspired oxygen. For most patients 10 litres/minute is sufficient oxygen flow through a reservoir mask. The oxygen flow is sufficient if the reservoir bag is not completely deflating.
- Manual ventilation bags should be reserved for patients requiring assistance with their breathing or requiring PEEP. For most patients 10 litres/minute is sufficient oxygen flow through a manual ventilation bag. The oxygen flow is sufficient if the reservoir bag is not completely deflating.
- Ensure a high degree of vigilance if a manual ventilation bag and mask is being held against the face of a small child who is spontaneously breathing, as a small child with respiratory exhaustion may not generate sufficient pressure to open the valves within the manual ventilation bag, and this may result in asphyxiation.

Pulse oximetry

- A pulse oximetry reading does not indicate how well a patient is breathing. How well a patient is breathing is determined by clinical examination and end tidal carbon dioxide ($ETCO_2$) measurement if this is available.
- Pulse oximetry can be unreliable if the patient is very vasoconstricted, shaking, moving, has very dirty fingers, or has been exposed to toxins such as carbon monoxide (CO) or nitrites.
- Do not spend long periods of time trying to obtain a pulse oximetry reading, noting that a common cause of failure to get a reading is severe vasoconstriction.
- Low pulse oximetry readings may be invalid if the SpO_2 waveform on the monitor is damped, flat or irregular. A wide and regular waveform is a sign that the pulse oximetry reading is likely to be valid.

Cyanosis

- A patient may be significantly hypoxic without being cyanosed because cyanosis is usually only detectable with an SpO_2 less than 80%.
- Cyanosis is much more difficult to detect in a patient who is anaemic, has a darker skin tone, or has been exposed to CO.

Oxygen Administration

- Central cyanosis (for example, of the mouth and lips) is usually due to severe hypoxia. Peripheral cyanosis (for example, of the extremities) in the absence of central cyanosis is usually due to vasoconstriction.

Oxygen administration and patients at risk of developing hypercarbia

- Some patients have carbon dioxide clearance that is altered by oxygen administration and excess oxygen administration in these patients may cause hypercarbia.
- Patients at high risk of hypercarbia include patients with COPD, morbid obesity, those on home oxygen and those on home CPAP or BiPAP.
- Oxygen flow rates should be titrated to the patient's known normal SpO_2 if this is known. If this is not known, titrate the oxygen flow rate to an SpO_2 of 88-92%.
- Bronchodilators should be nebulised using air if possible:
  - If using air to nebulise bronchodilators, administer bronchodilators using a flow rate of 8 litres/minute and administer oxygen via nasal prongs if required, adjusting the oxygen flow rate to achieve an SpO_2 of 88-92%.
  - If using oxygen to nebulise bronchodilators, alternating five minutes with the mask on and five minutes with the mask off should only occur if the SpO_2 climbs to above 92% during nebuliser delivery. This is done to limit oxygen exposure whilst delivering most of the nebulised bronchodilator.
- The signs of a rising carbon dioxide level are usually confusion, drowsiness, agitation and a falling level of consciousness. If a patient is suspected of developing hypercarbia, oxygen administration should not be discontinued immediately. Instead, oxygen administration should be reduced to a lower flow rate (targeting an SpO_2 of 88-92%) and the patient reassessed.

Oxygen administration and smoke inhalation

- Oxygen administration is not routinely required following smoke inhalation and should be reserved for patients that have suspected CO poisoning.
- See the **'poisoning from gases'** guideline for more information.

Oxygen administration following bleomycin treatment

- Oxygen administration can cause severe lung inflammation in patients previously treated with bleomycin (a chemotherapy drug) and should usually only be administered to maintain an SpO_2 of 88-92%.
- This sensitivity to oxygen is lifelong following exposure to bleomycin.

Oxygen Administration

Oxygen administration in confined spaces

- Use caution when administering oxygen in a confined space, for example, in a tank, pipe or silo.
- Oxygen administration in a confined space may lead to an increased oxygen concentration within the ambient air. An increase in the oxygen concentration within ambient air to as little as 24% may significantly increase the risk of fire and/or explosion.
- Only administer oxygen within a confined space if:
  - The clinical indication is very strong, and
  - FENZ personnel are present and are monitoring the oxygen concentration within the space.

PEEP (positive end expiratory pressure)

- PEEP improves oxygenation and reduces work of breathing via the following physiological effects:
  - The expiratory pressure reduces lung collapse.
  - The positive pressure reduces venous return to the heart and blood flow through lung vessels, reducing the amount of fluid entering the lungs.
  - The expiratory pressure increases the functional residual capacity. From this more expanded resting position, less work is required for inspiration because of the non-linear compliance of the lungs.
- In cardiac arrest:
  - PEEP is not applied to adults and children during CPR because it reduces the blood flow achieved during CPR. If ROSC is achieved it is appropriate to apply PEEP, but this is not an immediate priority.
  - PEEP is applied to neonates during CPR because the cause of cardiac arrest is usually respiratory failure and the balance of risk is in favour of improving ventilation.
- PEEP reduces cardiac output and should be used with caution in patients showing signs of shock.

More Information

Version history

16 June 2026 1.1.0.1
Update: Added PEEP guidance.

22 November 2022 1.0.1
Update: Indications for oxygen, oxygen administration threshold and guideline simplified.

Oxytocin

Mechanism of action

- Oxytocin is a synthetic version of the naturally occurring hormone oxytocin which is normally released from the pituitary gland.
- Oxytocin stimulates oxytocin receptors on the uterus, causing increased uterine contraction and reducing blood loss from the uterus.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Following normal birth.
- ✓ Postpartum haemorrhage.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Dosage

- For following normal birth: 10 units IM.
- For postpartum haemorrhage:
  - 10 units IV (preferred), or
  - 10 units IM.

Oxytocin

- If oxytocin IM has already been administered as part of routine treatment following normal birth, an additional dose of oxytocin should be administered (in the other thigh if administering IM) if postpartum haemorrhage develops. This may require meeting another vehicle.

Administration

- IV administration:
  - Place 10 units into a 1 litre bag of 0.9% sodium chloride. This solution contains 0.01 units/ml.
  - Shake well and label.
  - Administer IV over approximately 60 minutes (approx. 4 drops per second).
  - The rate of administration may be increased if shock is present.
- IM administration:
  - Administer IM undiluted.
  - The preferred site is the lateral thigh as this has the best absorption. If this site is not suitable use the lateral upper arm.
  - If multiple babies are present, administration must occur after delivery of the last baby.

Common adverse effects

- Abdominal cramping.
- Tachycardia.
- Flushing.

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 30-60 minutes.

Usual preparation

- Ampoule containing 10 units in 1 ml.

Oxytocin

Pharmacokinetics

- Oxytocin is metabolised in the liver and kidneys.
- There are no significant effects from liver or kidney impairment on acute administration.

Common interactions

- None.

Additional information

- Oxytocin that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is eight weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Routine administration of oxytocin following normal birth is controversial, but appears to reduce the incidence of postpartum haemorrhage.
- Oxytocin has been reported to cause prolongation of the QT interval if the patient is taking other medicines that also prolong the QT interval. However, this is usually associated with prolonged IV infusions and is not a clinically significant consideration when administering one or two IM doses.

More Information

Version history

9 June 2025 1.0.5.3

Update: Dosage and administration updated.

Paracetamol

Mechanism of action

- Paracetamol inhibits the production of prostaglandins resulting in a reduction in pain and fever.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.
- First Responders: refer to 'pain relief' in the FR Field Guide

Indications

- ✓ Mild or moderate pain.
- ✓ Moderate to severe pain, in addition to other medicines.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ The patient has taken paracetamol within the last four hours. Paracetamol is contained in many products such as cold and flu tablets/drinks, cough mixtures, combination pain relievers and migraine tablets. Additional paracetamol may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold paracetamol if there is any doubt.
- ⚠ Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting, the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- ⚠ Known severe liver disease. Liver disease must be severely impaired before paracetamol clearance is altered, but the balance of risk is such that paracetamol should usually be withheld in this setting.

Use in pregnancy or when breastfeeding

- Safe and may be administered if indicated.

Paracetamol

Dosage

- 1.5 g PO for an adult weighing greater than 80 kg.
- 1 g PO for an adult weighing 80 kg or less.
- See the paediatric drug dose tables for a child.
- **First Responders:** refer to '**pain relief**' in the FR Field Guide and administer the dose listed there.

Administration

- Administer PO.
- Children unable to swallow tablets may be administered paracetamol syrup, or tablets that have been crushed and placed in a soft food such as jam or honey.

Common adverse effects

- None.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 4-6 hours.

Usual preparation

- 500 mg tablets.
- Syrup containing 50 mg/ml.

Pharmacokinetics

- Paracetamol is metabolised in the liver.

Paracetamol

- If liver impairment is severe, paracetamol clearance will be significantly delayed, but this does not affect the initial (loading) dose.

Common interactions

- None.

Additional information

- Paracetamol should be administered in addition to other medicines for severe pain, particularly if the transport time is long. This is not a priority but will reduce the need for subsequent analgesia and improve the quality of pain relief.
- The described doses of paracetamol are higher than those usually recommended in many references. However, the described doses are safe provided they are not administered repeatedly.
- Paracetamol is not indicated for the treatment of fever because fever may confer some benefit if the patient has an infection. However, paracetamol may be administered if the patient has a temperature greater than 39°C and the fever is causing discomfort.
- Paracetamol is not indicated for pain associated with myocardial ischaemia.
- All personnel (including those without ATP) may give paracetamol to a patient for self-administration, provided the package instructions are followed.
- A patient may be administered paracetamol for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required.

More Information

Version history

22 November 2022 1.0.1

Update: Aligned with analgesia CPG.

Prednisone and Prednisolone

Mechanism of action

- Prednisone is a prodrug that is metabolised to prednisolone in the liver.
- Prednisolone is a corticosteroid with anti-inflammatory and immunosuppressant actions. It inhibits the production of inflammatory mediators, including prostaglandins and leukotrienes, resulting in a reduction in the inflammatory and immune response.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Bronchospasm associated with asthma or COPD.
- ✓ Croup.
- ✓ Minor allergy associated with rash.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ Age less than five years with asthma. Steroids do not usually have a role in children aged less than five years because they do not generally alter the course of their asthma exacerbation. However, a steroid is indicated if the child has a clear history of asthma and has previously received oral steroids.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. However, there is significant clinical experience with steroids and they appear to be safe. A steroid should be administered if there is bronchospasm, but should be withheld if the clinical problem is minor, for example rash or itch.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Prednisone and Prednisolone

Dosage

- 40 mg for an adult.
- See the paediatric drug dose tables for a child.
- If the patient is already taking prednisone:
  - Administer an additional full dose if the patient is taking a dose that is lower than that described in these CPGs. If the patient is not transported by ambulance to a medical facility, advise the patient to discontinue their usual prednisone, take the prednisone supplied by personnel and have their treatment reviewed in primary care (preferably by their own GP) within two days.
  - Do not administer an additional dose if the patient is taking a dose equal to or higher than that described in these CPGs. If the patient is not transported by ambulance to a medical facility, advise the patient to continue taking their usual prednisone and have their treatment reviewed by a doctor (preferably their own GP) within two days.

Administration

- Administer PO.
- Prednisone tablets are very bitter. Do not crush prednisone tablets because this may cause vomiting.
- The tablets may be divided. The tablet does not always break evenly, and this is not of clinical importance. The larger piece of the divided tablet should be chosen for administration.
- Patients aged greater than or equal to ten years should usually be administered prednisone tablets but may be administered the same dose of prednisolone syrup.
- Patients aged less than ten years should usually be administered prednisolone syrup but may be administered the same dose of prednisone tablets.
- Always ask a parent (or guardian) of a young child if the child can swallow tablets.

Common adverse effects

- Fatigue.
- Sodium and water retention. This may worsen hypertension and heart failure, but is usually only of clinical significance with prolonged dosing.
- Gastrointestinal reflux.

Prednisone and Prednisolone

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 24 hours.

Usual preparation

- 5 mg prednisone tablets.
- 20 mg prednisone tablets.
- Prednisolone syrup containing 5 mg/ml.

Pharmacokinetics

- Prednisone and prednisolone are predominantly metabolised by the liver.
- Prednisone is a prodrug that is metabolised to prednisolone in the liver. A dose of prednisone provides slightly less steroid equivalence than the same dose of prednisolone, but this is not clinically significant.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- None.

Providing the patient with prednisone

- If the patient is aged greater than or equal to 12 years and is not transported, additional prednisone should be provided to them by ambulance personnel, unless the patient already has an asthma/COPD action plan and their own supply of steroid.
- Provide the patient with an additional four days (four x 40 mg) of prednisone. Advise the patient and any carers, including in writing, that the patient:
  - Takes 40 mg (two 20 mg tablets) at approximately the same time each day, for the next four days until finished, in addition to their other usual medicines.

Prednisone and Prednisolone

» Should be reviewed in primary care for a review of their treatment within two days.

More Information

Version history

16 June 2026 1.1.0.1
Update: Treatment guidance updated.

28 October 2024 1.0.5
Update: Indications updated.

22 November 2022 1.0.1
Update: Prednisone pack information.

Rocuronium

Mechanism of action

- Rocuronium is a neuromuscular blocker. It antagonises (blocks) nicotinic acetylcholine receptors at the neuromuscular junction (motor nerve end plate) of skeletal muscle. This results in the inability of skeletal muscles to contract.

Authorised practice levels

- Only CCPs may administer rocuronium for the purposes of rapid sequence intubation (RSI).
- All ICPs may administer rocuronium post intubation, provided the endotracheal tube position has been confirmed with capnography.

Indications

- ✓ Neuromuscular blockade for RSI.
- ✓ Neuromuscular blockade following endotracheal intubation:
  - Rocuronium is routinely administered following RSI if suxamethonium is used as the neuromuscular blocker.
  - Rocuronium is administered if the patient shows clinically significant signs of moving following intubation without RSI.
- ✓ Patient movement during cardiac arrest that is interferring with resuscitation, despite ketamine administration, provided the patient has been intubated and the endotracheal tube position has been confirmed with capnography.

Contraindications

- ✕ Known severe allergy.
- ✕ Endotracheal tube position has not been confirmed by capnography (this does not apply if the indication is for RSI).

Rocuronium

Cautions

- ⚠ Predicted difficult intubation (this only applies if the indication is for RSI).
- ⚠ If the indication is for neuromuscular blockade following endotracheal intubation:
  - » Chronic muscle weakness. Examples include myasthenia gravis, motor neuron disease and muscular dystrophy. Rocuronium may cause prolonged muscle weakness and should be withheld if possible. If rocuronium is administered the dose should be halved.
  - » An adult with a very poor prognosis. Examples include severe comorbidities requiring long-term care, unwitnessed cardiac arrest and cardiac arrest with a first rhythm of asystole. Rocuronium should be withheld if possible because the patient is unlikely to benefit from admission to an intensive care unit and may be extubated in the ED.

Use in pregnancy or when breastfeeding

- › Safe and should be administered when indicated.

Dosage

- › For RSI: see the 'RSI' guideline.
- › For neuromuscular blockade following endotracheal intubation:
  - » 100 mg IV for an adult weighing greater than 80 kg.
  - » 50 mg IV for an adult weighing less than or equal to 80 kg.
  - » See the paediatric drug dose tables for a child.
  - » Repeat as required.

Administration

- › Administer IV as a bolus.

Common adverse effects

- › None.

Rocuronium

Usual onset of effect

- 30-60 seconds. This is predominantly affected by cardiac output and will be prolonged if cardiac output is low.

Usual duration of effect

- 30-60 minutes.

Usual preparation

- Ampoule containing 50 mg in 5 ml.

Pharmacokinetics

- Rocuronium is metabolised in the liver and excreted in urine.
- Significant hepatic or renal impairment will delay clearance and prolong the duration of effect.

Common interactions

- None.

Additional information

- Rocuronium that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is eight weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Rocuronium is a non-depolarising neuromuscular blocker. This means that the acetylcholine receptor at the neuromuscular junction is not stimulated prior to being blocked and no fasciculations will occur. This is in contrast to suxamethonium which is a depolarising neuromuscular blocker. Suxamethonium binds to the acetylcholine receptor at the neuromuscular junction and stimulates the receptor before blocking it, resulting in brief muscle contractions which are seen as fasciculations.

Rocuronium

- During cardiac arrest the blood flow generated by CPR may be insufficient to deliver adequate levels of rocuronium to skeletal muscle and neuromuscular blockade may not occur. If sustained ROSC is achieved, neuromuscular blockade will occur and the patient will require an adequate level of sedation.

- Adequate sedation and analgesia must always be administered to a patient if rocuronium is administered. See the **'post intubation'** guideline.

More Information

Version history

22 November 2022 1.0.1

Update: Discard date following removal from the fridge.

Salbutamol

Mechanism of action

- Salbutamol is a bronchodilator. It is an agonist (stimulator) of beta-2 receptors.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Bronchospasm secondary to asthma or COPD.
- ✓ Prominent bronchospasm secondary to airway burns, smoke inhalation or chest infection.
- ✓ Release syndrome following crush injury.
- ✓ Known or suspected hyperkalaemia with ECG changes.

Contraindications

- ✕ Known severe allergy.

Cautions

- ⚠ None.

Use in pregnancy or when breastfeeding

- Safe and may be administered if indicated.

Dosage

- MDI: One puff into the MDI and spacer, followed by six breaths. Repeat this six times.
- Nebulised for bronchospasm: 5 mg for adults and children, repeated as required. The initial dose is combined with 0.5 mg ipratropium, but subsequent doses are not.
- Nebulised for release syndrome following crush injury or known or suspected hyperkalaemia: 5 mg for adults and children, repeat as required.

Salbutamol

Administration

- MDI: Administer one puff of salbutamol using an MDI and spacer, followed by six breaths. Repeat this six times.
- Nebulised: Administer nebulised undiluted using:
  - Air as the driving gas for COPD.
  - Oxygen as the driving gas for all other indications.

Common adverse effects

- Tremor.
- Tachycardia.

Usual onset of effect

- 2-5 minutes.

Usual duration of effect

- 1-2 hours.

Usual preparation

- Metered dose inhaler delivering 100 mcg doses per puff.
- Ampoule containing 5 mg in 2.5 ml.
- Ampoule containing 2.5 mg in 2.5 ml.

Pharmacokinetics

- Only a small amount of the nebulised dose is absorbed, with most of the dose being nebulised to the atmosphere. The inhaled salbutamol is absorbed through the lungs and some is swallowed.
- Salbutamol is metabolised in the liver and excreted in urine.
- There are no significant effects from liver or kidney impairment on acute administration.

Salbutamol

Common interactions

- Salbutamol will be less effective in the presence of a beta-blocker, with the reduction in effect being most pronounced with a non-selective beta-blocker such as propranolol.

Additional information

- Salbutamol does not have a significant role in the treatment of bronchospasm as a result of smoke, toxic gas inhalation or chest infection. However, it may be administered if bronchospasm is prominent.

More Information

Version history

11 September 2023 1.0.4
Update: Indications, dosage and administration information updated.

22 November 2022 1.0.1
Update: Aligned with asthma and COPD CPGs.

Suxamethonium

Mechanism of action

- Suxamethonium is a neuromuscular blocker. It antagonises (blocks) nicotinic acetylcholine receptors at the neuromuscular junction (motor nerve end plate) of skeletal muscle. This results in the inability of skeletal muscles to contract.

Authorised practice levels

- CCPs.

Indications

- ✓ Neuromuscular blockade for rapid sequence intubation (RSI).

Contraindications

- ✕ Known severe allergy.
- ✕ Known personal or family history of malignant hyperthermia (MH). In susceptible patients suxamethonium will trigger life-threatening MH.
- ✕ Pre-existing paraplegia or quadriplegia. Long-term muscle weakness causes proliferation of acetylcholine receptors on skeletal muscle and suxamethonium may cause life-threatening hyperkalaemia.
- ✕ Muscle disorder with long-term weakness. Examples include muscular dystrophy and motor neurone disease. Long-term muscle weakness causes proliferation of acetylcholine receptors on skeletal muscle and suxamethonium may cause life-threatening hyperkalaemia.
- ✕ Hyperkalaemia is strongly suspected. Suxamethonium will cause a brief rise in the potassium concentration within blood of 0.5-1 mmol/litre. In the presence of hyperkalemia this rise may be significant.

Cautions

- ⚠ Predicted difficult intubation.

Use in pregnancy or when breastfeeding

- Safe and should be administered if indicated.

Suxamethonium

Dosage

- See the 'RSI' guideline.
- The onset of adequate neuromuscular blockade may be delayed if cardiac output is very low. Consider increasing the dose of suxamethonium to approximately 3 mg/kg, up to a maximum of 200 mg.

Administration

- Administer undiluted for an adult or a large child.
- Dilute 100 mg to a total volume of 10 ml for a small child. This solution contains 10 mg/ml.
- Administer IV as a bolus, preferably into a running IV line.

Common adverse effects

- Bradycardia. This is due to stimulation of acetylcholine receptors in the parasympathetic nervous system and is usually only clinically significant in young children.

Usual onset of effect

- 30-60 seconds. This is predominantly affected by cardiac output and will be prolonged if cardiac output is low.

Usual duration of effect

- 4-8 minutes.

Usual preparation

- Ampoule containing 100 mg in 2 ml.

Pharmacokinetics

- Suxamethonium is metabolised by the enzyme pseudocholinesterase. In most patients suxamethonium is cleared within 4-8 minutes.

Suxamethonium

- The neuromuscular blocking effects of suxamethonium can be increased to 4-12 hours if the patient has pseudocholinesterase deficiency.
- Pseudocholinesterase deficiency is rare.

Common interactions

- None.

Additional information

- Suxamethonium that is not within kits must be refrigerated:
  - Without refrigeration it loses approximately 5-10% of activity per month.
  - It may be safely stored in kits provided it is labelled with the discard date which is 8 weeks after removal from the fridge.
  - It may be transported for up to 12 hours between stores/stations in a container that keeps it cool, without affecting the expiry date, provided it is placed into a fridge immediately on arrival.
- Suxamethonium is a depolarising neuromuscular blocker. This means that the acetylcholine receptor at the neuromuscular junction is stimulated prior to being blocked and muscle fasciculations occur prior to the onset of neuromuscular blockade.
- Malignant hyperthermia (MH) is a rare, inherited disorder of muscle metabolism affecting approximately 20 families in New Zealand, many of whom are in the Manawatu area. When exposed to some anaesthetic agents (including suxamethonium) the patient may develop a life-threatening hypermetabolic state with severe hyperthermia. Patients with MH (or a family history of MH) usually know about it.
- Burn injury is often quoted as a contraindication to the administration of suxamethonium because life-threatening hyperkalaemia may occur. However, this is not a contraindication immediately following burn injury or once the patient is well enough to be discharged from hospital.

More Information

Version history

22 November 2022 1.0.1

Update: Discard date following removal from the fridge.

Tenecteplase

Mechanism of action

- Tenecteplase is a fibrinolytic that accelerates the breakdown of blood clots. It converts the plasma protein plasminogen into the active enzyme plasmin, which breaks down fibrin within blood clots.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ STEMI when following the fibrinolytic pathway.

Contraindications

- ✕ Known severe allergy.

Tenecteplase

Cautions

- More than ten minutes of CPR.
- Non-compressible vascular puncture within the last 24 hours.
- Internal bleeding within the last six weeks.
- Lumbar puncture or epidural insertion within the last six weeks.
- TIA within the last three months.
- Known bleeding disorder.
- Taking an anticoagulant.
- Systolic BP greater than 180 mmHg or diastolic BP greater than 110 mmHg.
- Known to be pregnant or less than two weeks postpartum.
- Time of onset of symptoms was greater than 12 hours ago.
- Requires assistance with their activities of daily living.
- Severe comorbidities that significantly limit their life expectancy.
- Suspected aortic dissection.
- Major surgery, major trauma or severe brain injury within the last six weeks.
- Intracranial surgery within the last six months.
- Ischaemic stroke within the last six months.
- Previous intracerebral haemorrhage.
- Known cerebral aneurysm, arteriovenous malformation or tumour.

Use in pregnancy or when breastfeeding

- Administration during pregnancy or within two weeks of birth carries a significant risk of bleeding. The likelihood of STEMI occurring in a woman who is pregnant or within two weeks of birth is very low and personnel must seek clinical advice prior to administration.

Tenecteplase

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- Dosage is based on age and known (or estimated) weight.

Weight	Age less than 75 years		Age 75 years or older	
	Tenecteplase (dose IV)	Tenecteplase (volume IV)	Tenecteplase (dose IV)	Tenecteplase (volume IV)
<60 kg	30 mg	6 ml	15 mg	3 ml
60-69 kg	35 mg	7 ml	17.5 mg	3.5 ml
70-79 kg	40 mg	8 ml	20 mg	4 ml
80-89 kg	45 mg	9 ml	22.5 mg	4.5 ml
≥90 kg	50 mg	10 ml	25 mg	5 ml

Administration

- Dissolve the powder using the syringe within the kit.
- Carefully discard unwanted drug from the syringe, preferably into the ampoule before administration, ensuring the correct dose remains in the syringe.
- If an error is made in discarding unwanted drug and the correct dose cannot be drawn up, administer the remaining drug, document this well, and notify the receiving clinician.
- Administer undiluted as an IV bolus, preferably into a running IV line.

Common adverse effects

- Bleeding. Tenecteplase commonly causes superficial bleeding, including epistaxis, bruising and bleeding from IV sites.
- Dysrhythmia. It is common for dysrhythmia to occur if the coronary artery reperfuses. Most commonly the rhythm is accelerated idioventricular rhythm (AIVR) which does not require specific treatment. Other dysrhythmias should be treated using the appropriate guideline.

Tenecteplase

Usual onset of effect

- 5-10 minutes.

Usual duration of effect

- 2-6 hours.

Pharmacokinetics

- Tenecteplase is metabolised by the liver.
- There are no significant effects from liver impairment on acute administration.

Usual preparation

- Glass ampoule containing 50 mg of tenecteplase, in powder form with a pre-filled syringe containing 10 ml of sterile water.

Common interactions

- None.

More Information

Version history

11 September 2023 1.0.4

Update: Indication wording has been changes. Cautions and controindications have been updated.

Ticagrelor

Mechanism of action

- Ticagrelor has antiplatelet activity. It produces reversible inhibition of the ADP receptor on platelets, decreasing platelet aggregation.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- STEMI, in conjunction with the primary PCI pathway, when requested by the accepting clinician.

Contraindications

- Known severe allergy.

Cautions

- Clinically significant bleeding. Ticagrelor will increase bleeding.
- At risk of bleeding.
- Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated during pregnancy. The likelihood of STEMI occurring in a woman who is pregnant is so low that personnel must seek clinical advice prior to administration.
- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 180 mg for an adult.

Ticagrelor

Administration

- Administer PO.

Common adverse effects

- Increased bleeding.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 24 hours.

Usual preparation

- 90 mg tablet.

Pharmacokinetics

- Ticagrelor is metabolised in the liver and excreted predominantly in the bile.
- There are no significant effects from liver impairment on acute administration.

Common interactions

- The risk of bleeding will be increased if the patient is taking an anticoagulant.

More Information

Version history

11 September 2023 1.0.4
New: Guideline released.

Tramadol

Mechanism of action

- Tramadol is an analgesic. It has multiple actions within the central nervous system, including opiate receptor stimulation and inhibition of the re-uptake of noradrenaline and serotonin.

Authorised practice levels

- EMTs, Paramedics, ICPs and CCPs.

Indications

- ✓ Moderate to severe pain, particularly if an opiate and/or ketamine is not going to be administered.

Contraindications

- ✕ Known severe allergy.
- ✕ Aged less than 12 years.

Cautions

- ⚠ Tramadol has been taken within the last four hours. Additional tramadol may be administered provided it is clear that the total dose taken within a four hour period does not exceed the dose described within the CPGs. Withhold tramadol if there is any doubt.
- ⚠ Abdominal pain, particularly if the patient is very unwell or vomiting. Clinical judgement is required, but if the patient is very unwell or vomiting, the possibility of significant intra-abdominal pathology exists and oral medicines should usually be withheld.
- ⚠ Aged greater than or equal to 75 years, particularly if there is a previous history of dementia or confusion. Tramadol has anticholinergic activity and this may cause confusion, particularly in the elderly.
- ⚠ Confusion. Tramadol has anticholinergic activity and this may worsen confusion.
- ⚠ Pregnancy.

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated in pregnancy and tramadol should usually be withheld.

Tramadol

- May be administered if the patient is breastfeeding. Advise the patient to stop breastfeeding and seek further advice from their lead maternity carer or GP.

Dosage

- 50 mg PO.

Administration

- Administer PO.

Common adverse effects

- Nausea and/or vomiting.
- Feeling light-headed or unusual.
- Sedation.
- Dry mouth.

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 4-8 hours.

Usual preparation

- 50 mg capsules.

Pharmacokinetics

- Tramadol is metabolised in the liver and excreted by the kidneys.
- There are no significant effects from liver or kidney impairment on the initial (loading) dose.

Tramadol

Common interactions

- Tramadol has been reported to cause serotonin syndrome in patients taking other medicines or recreational drugs that also raise serotonin levels within the brain. Examples include selective serotonin re-uptake inhibitors (SSRIs), tricyclic antidepressants and Ecstasy. However, this usually only occurs when doses of tramadol higher than 50 mg are taken regularly.

Additional information

- Tramadol is not indicated for pain associated with myocardial ischaemia.
- Do not routinely administer tramadol if an opiate or ketamine is administered because tramadol does not usually provide significant additional pain relief and may worsen side effects. However, tramadol may be administered if the patient has taken tramadol before and found it effective.
- Some patients experience nausea and/or feel unusual with tramadol and may refuse to have it again. These are side effects, not an allergy and are most likely to occur with doses higher than 50 mg administered IV.
- Tramadol has been reported to lower the seizure threshold in patients with epilepsy. However, this usually only occurs with doses higher than 50 mg taken regularly.
- A patient may be administered tramadol (in combination with paracetamol and/or ibuprofen) for pain associated with a minor condition and be given a recommendation that immediate referral to a medical facility is not required provided:
  - Adequate pain control is achieved, and
  - The patient is advised to be seen in primary care for a review of their condition.

More Information

Version history

22 November 2022 1.0.1

Update: Aligned with analgesia CPG and caution added.

Tranexamic Acid

Mechanism of action

- Tranexamic acid is an antifibrinolytic medicine. It blocks the conversion of plasminogen to plasmin, reducing fibrinolysis (breakdown of blood clots) and bleeding.

Authorised practice levels

- Paramedics, ICPs and CCPs.

Indications

- ✓ Postpartum haemorrhage.
- ✓ Clinically significant bleeding or signs of hypovolaemia following trauma.
- ✓ Non-traumatic bleeding and shock is severe.
- ✓ Cardiac arrest secondary to trauma.
- ✓ Bleeding following tonsillectomy.
- ✓ Crush injury.
- ✓ Moderate to severe epistaxis, not controlled following one dose of adrenaline IN.
- ✓ Antepartum haemorrhage if shock and/or bleeding is severe.

Contraindications

- ✕ Known severe allergy.
- ✕ Trauma when tranexamic acid will be administered more than three hours after the time of injury. This appears to increase mortality rates, the cause of which is not clear.

Cautions

- ⚠ None.

Tranexamic Acid

Use in pregnancy or when breastfeeding

- Safety has not been demonstrated, but the balance of risk is such that it should be administered if indicated.

Dosage

- For bleeding following tonsillectomy:
  - 500 mg nebulised for an adult or child. Repeat once after 30 minutes.
  - 1 g IV for an adult. See the paediatric drug dose tables for a child.
- For bleeding following trauma:
  - 2 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- Non-traumatic bleeding:
  - 1 g IV for an adult.
  - See the paediatric drug dose tables for a child.
- For moderate to severe epistaxis:
  - Administer 200 mg (2 ml undiluted) per bleeding nostril for a patient aged 12 years or over.
  - Administer 100 mg (1 ml undiluted) per bleeding nostril for a child aged 5-11 years.
  - Seek clinical advice if the patient is a child aged less than five years.

Administration

- IV: administer over 1-2 minutes.
- Nebulised: administer undiluted.
- IN: administer undiluted.

Common adverse effects

- None.

Tranexamic Acid

Usual onset of effect

- 30-60 minutes.

Usual duration of effect

- 8 hours.

Usual preparation

- 10 ml ampoule containing 1 g.
- 5 ml ampoule containing 500 mg.

Pharmacokinetics

- Tranexamic acid is predominantly excreted in the urine.
- Renal impairment does not alter the initial (loading) dose.

Common interactions

- None.

Additional information

- Some references describe the topical administration of tranexamic acid for control of superficial bleeding. This usually has no role in the out-of-hospital setting and personnel should seek clinical advice if it is being considered.

More Information

Version history

9 June 2025 1.0.5.3
Update: Indications and dosage updated.

28 October 2024 1.0.5
Update: Indications, dosage and route updated.

Tranexamic Acid

11 September 2023 1.0.4

Update: Indications have been updated. Reference to PATCH trial removed.